

Anticipating demand in carbon-aware scheduling across data centres: guarantees and fair carbon attribution

Quang-Vinh Dang^{1*}

^{1*}British University Vietnam, Hung Yen, Vietnam.

Corresponding author(s). E-mail(s): vinh.dq4@buv.edu.vn;

Abstract

Shifting deferrable jobs to cleaner hours and locations cuts data-centre emissions, but online schedulers act before future grid conditions and jobs are known. We formulate multi-site, deadline-constrained scheduling of bag-of-tasks jobs as a minimum-cost flow problem and propose CARMA (Carbon-aware Anticipatory Receding-horizon Multi-site Allocation), which adds “ghost jobs” for expected arrivals to each hourly plan. We prove that uncertainty about demand alone forces any online algorithm, even one with perfect carbon forecasts, to a competitive ratio of $\Omega(\sqrt{\theta})$, where θ is the ratio of the highest to the lowest unit emission, and myopic planning to at least $(1 + \theta)/2$. Given backstop capacity, CARMA never misses a deadline under any predictions, is optimal under exact predictions and degrades linearly with prediction error. Dual prices of the pooled problem yield a carbon attribution among sites that lies in the core. On 2024 regional grid data for Great Britain, CARMA comes within 8.0–9.8% of the clairvoyant optimum, against 15–16% for the best myopic, protection-level or risk-averse baseline and 19–23% for plain model-predictive control, and cuts attributional emissions by 36–68% relative to carbon-agnostic operation. On a real cluster trace with a demand profile learned from different days, its gap is 20.7% against 26.7–36.1%. Anticipating demand is worth more than perfect carbon foresight. Under stylised marginal-emission accounting, achievable reductions shrink to 14% and CARMA’s edge becomes small, so physical savings depend on curtailed low-carbon supply. Origin- and host-based carbon accounting violate coalitional stability in every week, whereas the dual attribution never does in hindsight.

Keywords: Carbon-aware computing, Data centre scheduling, Competitive analysis, Minimum-cost flow, Cooperative cost allocation, Carbon intensity forecasting

047 1 Introduction

048
049 Data centres are among the fastest-growing electricity consumers. Efficiency gains
050 kept their energy use nearly flat through the 2010s even as computing output grew
051 several-fold [1, 2]. That buffer is now thinning, and information and communication
052 technology accounts for an estimated 2.1–3.9% of global greenhouse-gas emissions [3].
053 Large machine-learning workloads have drawn particular attention [4–6]. The carbon
054 intensity of grid electricity varies by a factor of ten or more across regions and hours,
055 driven by the output of wind, solar and dispatchable plant [7]. Operators can therefore
056 cut emissions without new hardware by moving flexible work to cleaner places and
057 times. Such *carbon-aware computing* [8, 9] is an operations problem: capacity is finite,
058 deadlines bind, and both the grid and the job stream are uncertain.

059 Batch jobs such as model training, analytics, rendering and backups are a natural
060 target, because they tolerate some delay. Previous work has shown the potential of
061 temporal shifting (running later) [10–12] and of spatial shifting (running elsewhere) [13,
062 14]. It has also shown their limits. Sukprasert et al. [15] found that practical reductions
063 from spatio-temporal shifting fall well short of idealised bounds. Most methods are
064 evaluated with a scheduler that reacts to the jobs it has already seen, using point
065 forecasts of carbon intensity. Three questions have received less attention. First, how
066 much is lost because an online scheduler does not know which jobs will arrive next,
067 and can any scheduler guarantee good performance in the worst case? Second, does it
068 pay to price the uncertainty of carbon-intensity forecasts into scheduling decisions?
069 Third, when several sites pool their capacity, how should the resulting emissions be
070 attributed so that every participant is better off than on its own?

071 This paper addresses these questions with a scheduling policy, guarantees under
072 explicitly stated assumptions, and a controlled evaluation on three years of regional
073 grid data for Great Britain. Its building blocks (receding-horizon planning, network
074 flows, conformal calibration and dual cost allocation) are standard; the contribution
075 lies in their combination for this problem, the lower bound, the conditions under which
076 the guarantees hold and the empirical evidence. It makes five contributions.

- 077 1. **Formulation.** We state multi-site, deadline-constrained, width-limited scheduling of
078 deferrable bag-of-tasks jobs, including the emissions of data transfer, as a minimum-
079 cost flow problem (Proposition 1). Offline clairvoyant optima and every online
080 planning step can therefore be solved to optimality, after quantisation, by fast
081 combinatorial min-cost flow algorithms. We also prove that myopic receding-horizon
082 scheduling can be arbitrarily worse than the clairvoyant optimum even with perfect
083 carbon forecasts (Proposition 2).
- 084 2. **Algorithm.** We propose CARMA (Carbon-aware Anticipatory Receding-horizon
085 Multi-site Allocation). At every hour, CARMA adds *ghost jobs* to its planning
086 network. These represent expected arrivals, learned from the operator’s history, so
087 the planner reserves scarce low-carbon capacity for work that has not yet arrived. The
088 planning problem stays a single min-cost flow and decisions take tens of milliseconds.
- 089 3. **Guarantees.** We prove that uncertainty about demand alone forces every online
090 algorithm, even one with perfect carbon forecasts, to a competitive ratio of $\Omega(\sqrt{\theta})$,
091 about $\sqrt{\theta}/2$ for large θ , where θ is the ratio of the highest to the lowest unit emission
092

(Theorem 1). In an idealised setting with a backstop resource, CARMA never misses a deadline whatever its predictions, so it inherits the θ -competitiveness that every on-time policy has. If, in addition, each episode fits within the planning window, CARMA is optimal when its predictions of demand and carbon are exact, and its excess emissions grow at most linearly in the prediction errors (Proposition 3 and Theorem 2). These assumptions do not hold in our weekly experiments, so we verify the theorem separately in episodes that satisfy them. We also show that the dual prices of the pooled scheduling network give a carbon attribution among sites that lies in the core of the pooling game, so no group of sites gains by leaving (Theorem 3).

4. **Forecasting and recalibration pipeline.** As a tool for the scheduling study, we build a reproducible 1–24-hour carbon-intensity forecaster from a public feed: direct gradient-boosted models with horizon-wise normalised conformal quantiles and adaptive recalibration, which keeps empirical coverage close to target under the decarbonisation drift of 2024. Its quantiles let us test risk-adjusted carbon pricing inside the scheduler. We do not claim a new forecasting method.

5. **Evidence.** Across 52 out-of-sample weeks with three workload traces each, three load levels, two operating regimes and a trace-driven workload from the Alibaba 2018 cluster trace, we compare CARMA with seven implementable online baselines, including a revenue-management protection-level policy and a scenario-based stochastic lookahead that uses the same demand information, with clairvoyant and ablated variants, and with the clairvoyant oracle. The results separate the value of anticipating demand from the value of better carbon forecasts, show when each matters, characterise how performance degrades with prediction errors, and show that origin- and host-based carbon accounting violate coalitional stability in every test week, whereas the dual attribution satisfies it exactly for the hindsight benchmark and in over 99% of the checks for CARMA’s realised emissions.

The results carry an operational message. Once spatial shifting is allowed, the dominant loss comes from demand myopia, not from forecast error. Anticipating demand cuts the gap to the clairvoyant optimum by more than half in the synthetic experiments, and by 43% on a real cluster trace whose demand profile is learned from different days. Perfect carbon foresight or risk-averse carbon pricing adds little by comparison. Without migration, temporal shifting alone achieves much smaller reductions, and near saturation it trades emissions against lateness. We also show the limits of the evidence. The headline reductions are attributional. Under a stylised marginal-emission accounting within the single GB market, most of them are reallocations, and physical savings depend on low-carbon supply that would otherwise be curtailed. The rest of the paper is organised as follows. Section 2 reviews related work. Section 3 presents the model, the forecaster, CARMA and the experimental design. Section 4 establishes the theoretical guarantees, with proofs in Appendix A. Section 5 reports results, Section 6 discusses implications and limitations, and Section 7 concludes.

2 Related work

Carbon-aware workload shifting

Early green data-centre schedulers matched deferrable jobs to on-site solar production, for example GreenSlot [16] and GreenHadoop [11]. Other work exploited electricity-price and carbon differences across sites through geographical load balancing [13, 17] and dynamic right-sizing [18]. With public carbon-intensity feeds, attention moved to grid emissions. Wiesner et al. [10] quantified the reductions from temporal shifting in several regions, including Great Britain, and proposed forecast-based lowest-window scheduling. Google’s carbon-intelligent compute system sets next-day virtual capacity curves from carbon forecasts and forecasts of the fleet’s own demand [8]. It is the closest industrial precedent for using demand forecasts, but it shapes hourly virtual capacity curves per cluster rather than planning individual jobs across sites. Cucumber admits delay-tolerant work only when forecast renewable supply allows it [19], and Caribou shifts serverless functions between regions [20]. CarbonScaler varies the allocation of elastic jobs with carbon intensity [12]. Ecovisor exposes the energy system to applications [21]. Carbon Explorer weighs scheduling against investment in renewables and storage [22]. Load migration between data centres can absorb curtailed renewable generation [14], and data-centre demand response has been surveyed in [23]. Sukprasert et al. [15] bound the potential of spatio-temporal shifting across 123 regions and find that simple policies capture most of the achievable benefit. Our study complements theirs by identifying *which* information limits online schedulers when capacity at clean sites is scarce.

Online algorithms and guarantees

Worst-case analysis of carbon-aware scheduling builds on online search and one-way trading [24], for which threshold policies achieve optimal competitive ratios. Lechowicz et al. derived optimal double-threshold algorithms for the online pause-and-resume problem [25]. With CarbonClipper, they also derived learning-augmented algorithms with an optimal consistency–robustness trade-off for spatiotemporal allocation of a single deadline-constrained workload over a network of data centres [26]. These works model uncertainty about future carbon *prices* for one job. We study many jobs that compete for finite low-carbon capacity, and we show that uncertainty about *demand* creates a distinct lower bound that carbon forecasts cannot remove. Our guarantees follow the consistency–robustness paradigm of algorithms with predictions [27]. CARMA’s re-solving of a deterministic plan with expected future demand is an instance of the certainty-equivalent re-solving heuristics of network revenue management [28]. Our contribution is not the device itself but its formulation as a single flow problem for carbon-aware scheduling, the guarantees and the empirical attribution of value. Table 1 positions the closest work along the dimensions that matter here.

Our fairness result draws on cooperative game theory: linear production games and flow games have non-empty cores that can be computed from dual prices [31, 32].

Table 1 Positioning against the closest work, as modelled in the cited studies (✓: explicitly modelled or provided; −: not modelled).

Work	Many jobs share finite capacity	Information on future demand	Worst-case guarantee	Attribution among
Lowest-window temporal shifting [10]	−	−	−	−
Geographical load balancing [13]	✓	−	−	−
Carbon-intelligent compute system [8]	✓	✓	−	−
Online pause-and-resume [25]	−	−	✓	−
CarbonClipper [26]	−	−	✓	−
Flow models of preemptive scheduling [29, 30]	✓ (offline)	−	−	−
This paper	✓	✓	✓ (A1–A2)	✓

Energy-aware scheduling in operations

Energy-efficient scheduling is a mature topic in manufacturing operations [33], and green scheduling models have appeared in this journal, for example for parallel machines with job splitting [34] and for cloud resource allocation [35]. Most such models are deterministic and solved offline, often with metaheuristics. We instead exploit exact network-flow structure and embed it in an online lookahead policy. Network-flow formulations of preemptive scheduling with release dates and deadlines go back to Horn [29] and Federgruen and Groenevelt [30]. Proposition 1 extends this classical construction to multiple sites, job widths, time-varying emission costs and transfer emissions [36, 37]. In the taxonomy of Powell [38], CARMA is a deterministic lookahead policy with a modified (anticipatory) model, in the spirit of model predictive control [39, 40].

Carbon-intensity forecasting

Forecasts of grid carbon intensity use statistical decomposition [41], machine learning with many exogenous drivers [42], day-ahead regression for building demand response [43], and hierarchical models of the generation mix, as in CarbonCast [44]. For Great Britain, NESO publishes regional forecasts through its Carbon Intensity API [45]. These are average (attributional) intensities. Marginal emission factors, which describe the plant that responds to a change in load, can differ substantially from averages and are the relevant measure for the system-level effect of load shifting [46, 47]. These studies focus on point accuracy. For decisions, calibrated uncertainty matters as well. Conformal prediction [48–50] provides distribution-free intervals and has been extended to quantile regression [51], to multi-horizon [52] and dynamic time series [53], and to arbitrary distribution shift through adaptive conformal inference [54]. We use these tools to build a recalibrated forecaster and to test, in a controlled way, whether risk-adjusted carbon costs improve scheduling. The question is motivated by the optimizer’s curse [55] and by the robust-optimisation view that point-estimate solutions can be fragile [56, 57].

3 Material and methods

3.1 Problem setting

We consider an operator that runs a set $\mathcal{S}$ of data-centre sites connected by a wide-area network. Time is divided into hourly slots $t = 0, 1, 2, \dots$. Deferrable batch jobs arrive online. Job j is released at the start of hour a_j at origin site $o_j \in \mathcal{S}$. It carries w_j server-hours of work made of independent tasks (a bag of tasks), can use at most m_j servers concurrently, and must finish before the start of hour $d_j > a_j$. Work may be paused and resumed, and any part of it may run at a site other than its origin. Remote execution requires moving input data. At site s and hour t , $K_s(t)$ servers are free for batch work once the interactive load has been served.

Let $I_s(t)$ be the average carbon intensity of the electricity supplied to site s during hour t (gCO₂/kWh). Executing one server-hour of job j at site s during hour t emits

$$c_{o_j s}(t) = e \text{PUE}_s I_s(t) + \mathbf{1}[s \neq o_j] \delta \eta \frac{1}{2} (I_{o_j}(t) + I_s(t)), \quad (1)$$

where e is the IT energy per server-hour, PUE_s is the power usage effectiveness of site s , δ is the data volume moved per migrated server-hour and η is the energy intensity of network transfer (kWh/GB). The transfer term charges network energy at the mean intensity of the two endpoints. With decision variables $x_{js}(t) \geq 0$ (server-hours of job j run at site s in hour t), the offline version of the problem is the linear programme

$$\min_{x, u \geq 0} \quad \sum_j \sum_s \sum_{t=a_j}^{d_j-1} c_{o_j s}(t) x_{js}(t) + \sum_j \Pi u_j \quad (2)$$

$$\text{s.t.} \quad \sum_s \sum_{t=a_j}^{d_j-1} x_{js}(t) + u_j = w_j \quad \forall j, \quad (3)$$

$$\sum_s x_{js}(t) \leq m_j \quad \forall j, t \in [a_j, d_j), \quad (4)$$

$$\sum_{j: a_j \leq t < d_j} x_{js}(t) \leq K_s(t) \quad \forall s, t, \quad (5)$$

where u_j is work left unfinished at the deadline and Π is a lateness penalty much larger than any emission cost. When migration is not allowed, $x_{js}(t) = 0$ for $s \neq o_j$; these variables are removed from the problem. With complete knowledge of arrivals and intensities, the optimum of (2)–(5) is a lower bound on the emissions of every schedule that serves all jobs by their deadlines, and we use it as the *clairvoyant oracle*. When emissions are evaluated only for a subset of jobs (Section 3.5), the oracle still has to serve every job under the same constraints but is charged emissions only for the evaluated ones, so that it remains a lower bound on the evaluated emissions of any schedule that meets all deadlines. Online policies that finish some work late lie outside this set; Section 3.5 explains how we compare them. An online scheduler has to fix $x_{js}(t)$ at the start of hour t . At that point it knows the jobs released so far, the current intensities $I_s(t)$ and nothing more than forecasts of $I_s(t+h)$ and of future arrivals.

Proposition 1 (Network-flow structure) *Problem (2)–(5) is a minimum-cost flow problem on a network with a source, one node per job, one node per (job, hour) pair, one node per*

(hour, site) pair and a sink. The source supplies $\sum_j w_j$ units of flow and the sink absorbs them. Arc capacities are w_j (source $\rightarrow j$), m_j ($j \rightarrow (j, t)$) and $K_s(t)$ ($(t, s) \rightarrow \text{sink}$). The only arcs with a cost are $(j, t) \rightarrow (t, s)$, priced at $c_{ojs}(t)$, and a slack arc $j \rightarrow \text{sink}$ priced at Π .

Proof Because the supply equals $\sum_j w_j$ and arc source $\rightarrow j$ has capacity w_j , every such arc is saturated, and conservation at the job node is Eq. (3). The capacity of arc $j \rightarrow (j, t)$ is Eq. (4), conservation at (j, t) splits that hour's work across sites, and the capacity of $(t, s) \rightarrow \text{sink}$ is Eq. (5). Conversely, every feasible flow defines a feasible (x, u) with the same cost. The constraint matrix of the flow formulation is a node-arc incidence matrix, in which every column has exactly one +1 and one -1, so it is totally unimodular. $\square$

Proposition 1 has a practical consequence. The oracle and every planning problem solved online (Section 3.3) can be solved by combinatorial min-cost flow algorithms [36], which return integral optimal flows for integral data. We quantise work, widths and capacities to 0.01 server-hours (widths rounded up, capacities down) and costs to 0.01 gCO₂, and solve the quantised network to optimality with the min-cost flow solver of OR-Tools [58] (version 9.15, a cost-scaling push-relabel algorithm). The solutions are therefore optimal for the quantised problem, and the simulator treats remaining work below 0.05 server-hours as complete; it checks every executed allocation against the capacity and width limits up to this tolerance. On the weekly clairvoyant problems (1,100–2,800 jobs), the flow solver returned the same objective as the LP solver HiGHS on the unquantised problem to within a relative 5×10^{-5} and was 3–8 times faster (script `bench_solvers.py`; one CPU core).

3.2 Probabilistic carbon-intensity forecasting

Point model

For each horizon $h = 1, \dots, H$ ($H = 24$ h) we fit a direct gradient-boosted regression tree model [59] that predicts the change $I_s(t+h) - I_s(t)$ from information available at issue time t . Direct per-horizon models avoid the error accumulation of recursive strategies [60]. One model per horizon is pooled across sites, with the site as a categorical feature. The features are the current intensity; differences between the current value and lags of 1, 2, 3, 6, 12 and 23 hours; the value observed at the same hour of the target on the previous day and in the previous week; a 24-hour rolling mean and mean absolute change (volatility); the national (GB) intensity; the regional wind, solar and gas shares; and calendar encodings of the target time (hour of day, day of week, annual sine and cosine). Models are fitted with absolute-error loss, so they estimate the conditional median. We use scikit-learn's histogram gradient boosting [61] with 400 iterations, learning rate 0.05, 31 leaves and at least 50 samples per leaf. The models use only information available when the forecast is issued (gaps in the data are filled causally; Section 3.5), and no weather forecasts. This is deliberate: the goal is a forecaster any operator can build from a public intensity feed.

Conformal upper quantiles

A point forecast $\hat{I}_s(t+h|t)$ is turned into calibrated upper quantiles by split conformal prediction [48, 49] with normalised scores. On a calibration window disjoint from

323 training, we compute for each site s and horizon h

324

$$325 \quad r_{s,h}(t) = \frac{I_s(t+h) - \hat{I}_s(t+h|t)}{\sigma_s(t) + \epsilon}, \quad (6)$$

326

327 where $\sigma_s(t)$ is the 24-hour mean absolute hourly change observed at issue time and
 328 $\epsilon = 5 \text{ gCO}_2/\text{kWh}$. Only pairs whose target $t+h$ also lies in the calibration window are
 329 used, so that no calibration outcome falls in the tuning period. For a target level β ,
 330 the upper forecast is $U_s^\beta(t+h|t) = \hat{I}_s(t+h|t) + \hat{q}_{s,h}(\beta, \sigma_s(t)) (\sigma_s(t) + \epsilon)$, where $\hat{q}_{s,h}(\beta)$
 331 is the $\lceil (n+1)\beta \rceil$ -th smallest of the n calibration scores, the split-conformal quantile.
 332 Grid carbon intensity is not exchangeable over time: 2024 was markedly cleaner than
 333 2022–2023 (Fig. 3). We therefore update the working level online by adaptive conformal
 334 inference (ACI) [54]. Once $I_s(t+h)$ is observed,

335

$$336 \quad \alpha_{s,h} \leftarrow \alpha_{s,h} + \gamma((1-\beta) - \mathbf{1}[I_s(t+h) > U_s^\beta(t+h|t)]), \quad \beta_{s,h}(t) = 1 - \alpha_{s,h}, \quad (7)$$

337

338 with step size $\gamma = 0.005$. At every hour, the outcomes observed in that hour update
 339 the levels before the new forecasts are issued, so feedback for horizon h arrives h hours
 340 after the forecast. When the working level reaches 1, the bound is set to the largest
 341 intensity of the site in the training period, a finite worst case in place of the infinite
 342 interval of ACI; when it reaches 0 the bound is zero, and bounds are floored at zero.
 343 Gap-filled outcomes give no feedback. For immediate feedback and unbounded intervals,
 344 ACI guarantees that the long-run miscoverage frequency converges to $1 - \beta$ for any
 345 data sequence [54]. Our variant has delayed feedback, a finite ceiling and a quantile
 346 grid of step 0.001, so we do not claim this guarantee and use ACI as an adaptive
 347 recalibration whose coverage is checked empirically (Appendix B). The level β then
 348 acts as a risk dial that remains roughly calibrated under drift, and with $\beta = 0.5$ the
 349 procedure removes level shifts in the point model (a recalibrated median).
 350

351

352 3.3 CARMA: carbon-aware anticipatory receding-horizon 353 multi-site allocation

354

355 A standard receding-horizon (model predictive control, MPC) scheduler [39] solves,
 356 at every hour t , the flow problem of Proposition 1 restricted to pending jobs and the
 357 window $[t, t+H)$. Future intensities are replaced by forecasts, and only the first hour
 358 of the plan is executed. Such a planner is *myopic about demand*: it sees only jobs that
 359 have already arrived. The next result shows that this can be arbitrarily costly even
 360 when carbon intensities are known exactly.

361

362 **Proposition 2** (Myopia can be unboundedly costly) *Even with perfect intensity forecasts,*
 363 *myopic receding-horizon scheduling has no finite competitive ratio with respect to the clairvoyant*
 364 *oracle.*

365

366

367 *Proof* Take a clean site A with one server and a dirty site B with ample capacity. Let $c_A(0) = \epsilon$,
 368 $c_A(1) = 0$ and $c_B(0) = c_B(1) = c$, and ignore transfer costs. A flexible job ($w = 1$, $m = 1$)

arrives at $t = 0$ with deadline $d = 2$. An urgent job ($w = 1$) arrives at $t = 1$ with $d = 2$. At $t = 0$ the myopic planner sees only the flexible job and defers it to A at $t = 1$ (cost $0 < \varepsilon$). At $t = 1$ both jobs must run in the same hour, one of them at B , so the total cost is c . The oracle runs the flexible job at $t = 0$ and the urgent job at $t = 1$, both at A , for a total of ε . The ratio c/ε is unbounded as $\varepsilon \rightarrow 0$. $\square$

The mechanism in Proposition 2 is common in practice. Low-carbon capacity is scarce, and a myopic planner spends it on flexible work that could have waited or run elsewhere, leaving urgent jobs that arrive later to run on carbon-intensive capacity. CARMA addresses it by anticipating demand inside the same flow problem, and optionally prices forecast risk.

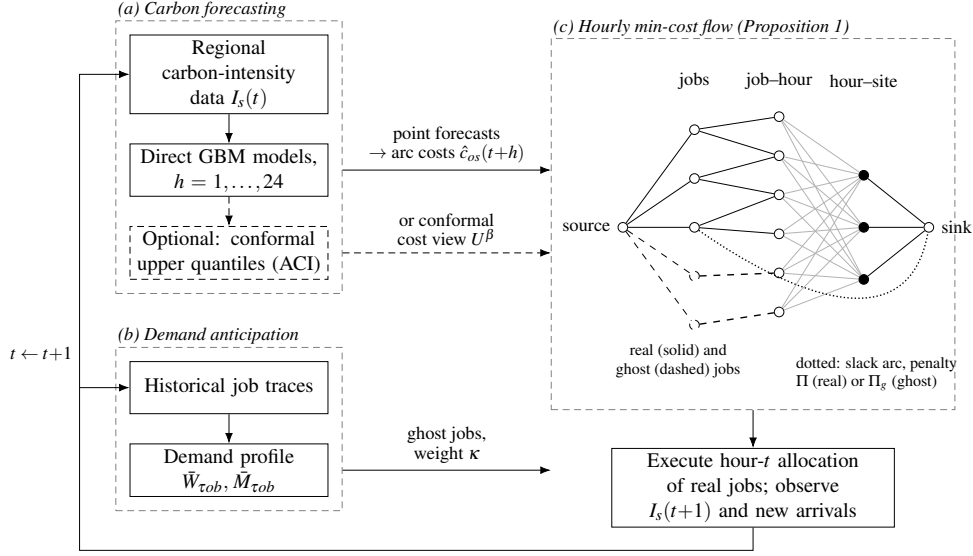


Fig. 1 The CARMA decision loop. (a) Carbon-intensity forecasts price the job-hour to hour-site arcs; the selected configuration uses point forecasts, and conformal upper quantiles are an optional cost view. (b) A demand profile learned from past traces generates ghost jobs for expected arrivals. (c) Real jobs (solid) and ghost jobs (dashed) share one min-cost flow network, whose dotted slack arc carries unfinished work to the sink at the lateness penalty. The network is solved exactly every hour, and only the first-hour allocation of real jobs is executed.

Anticipated demand (ghost jobs)

From the operator's own history we estimate, for every hour of the week τ , origin o and window class b , the expected work $\bar{W}_{\tau ob}$ and the expected total width $\bar{M}_{\tau ob}$ of arriving jobs. Window classes are the bins $\mathcal{B} = \{1, 2, 4, 6, 9, 13, 18, 24\}$ hours. A job is assigned to the largest bin not exceeding its window, so ghost jobs are never more flexible than the jobs they represent. Because binning shortens windows, the width of each ghost job is raised where necessary to at least its work divided by its window, which makes every ghost job window-feasible. At hour t , for each $k = 1, \dots, \ell$ (the look-ahead), CARMA

adds to the planning problem a *ghost job* for every (o, b) with work $\kappa \bar{W}_{\tau(t+k)ob}$, width $\kappa \bar{M}_{\tau(t+k)ob}$, release $t + k$ and deadline $\min(t + k + b, t + H)$. Ghost jobs compete with real jobs for capacity, but their slack arc is priced at a moderate penalty Π_g , set to 1.5 times the largest local unit cost in the window, rather than at Π . They can therefore reserve clean capacity when it is economical but never force real work to be late. The weight $\kappa \geq 0$ scales how much of the expected demand is anticipated. With $\kappa = 0$, CARMA reduces to myopic MPC.

Carbon cost view

Future-hour costs in Eq. (1) are evaluated at a forecast of $I_s(t + h)$; for the current hour ($h = 0$) the observed intensity is used. CARMA admits two cost views: the point forecast $\hat{I}$, or the conformal upper quantile U^β of Section 3.2. The second is motivated by the optimizer’s curse [55]: selecting the minimum over noisy estimates is biased towards optimism. The conformal view replaces each future cost by a calibrated quantile, which shifts it by a site- and horizon-specific amount proportional to the local volatility. For $\beta > 0.5$ this penalises volatile sites and distant hours more than stable sites and near hours. Whether this risk premium helps is an empirical question, and we treat the cost view as a hyperparameter.

Algorithm

Fig. 1 summarises the loop. At each hour t , CARMA (i) observes $I_s(t)$, the free capacity and the pending jobs; (ii) builds the cost cube $c_{os}(t + h)$ for $h < H$ from the chosen forecast view; (iii) adds the ghost jobs for hours $t + 1, \dots, t + \ell$; (iv) solves the resulting min-cost flow problem exactly; (v) executes the first-hour allocations of the real jobs only; and (vi) after observing the realised intensities, updates the ACI levels by Eq. (7) when conformal costs are used. Work that is still pending at its deadline receives a one-hour planning window, which makes it urgent in the next plan; it carries the same penalty as other real work, so no strict priority is imposed. The planning network has $O((n_t + \ell|\mathcal{S}||\mathcal{B}|)H|\mathcal{S}|)$ arcs, where n_t is the number of pending jobs, so each decision takes a few tens of milliseconds (Section 5). The hyperparameters (κ, ℓ) and the cost view are tuned on validation weeks that precede the test period (Section 5.2).

3.4 Baselines

We compare CARMA with the following online policies and the oracle. **ASAP-Local** is carbon-agnostic: it runs work at its origin as soon as possible, earliest deadline first (EDF). **Greedy-Spatial** also runs work immediately, but at the site that is currently cleanest; this is geographical load balancing without temporal shifting [13]. **Forecast-Reserve** books, when a job arrives, the cheapest forecast (site, hour) slots in its window that still have capacity, and never revises the booking. This is the lowest-window strategy of [10] extended to multiple sites. **MPC** is the myopic receding-horizon flow scheduler with point forecasts. **MPC-Protect** adds a protection level in the spirit of revenue management: planned work may use only a share $1 - r$ of the capacity of the sites that are cleaner than the fleet median over the window. In the current hour, the protected share is first offered to jobs due within τ hours and then, because unused capacity perishes, to all remaining work. The parameters (r, τ) are tuned on validation

data. **MPC+conformal** uses the conformal upper quantile at the level that was best for MPC on validation data. **MPC-Perfect** is MPC given the realised intensities, which isolates the value of forecast accuracy. **CARMA-Perfect** is CARMA given the realised intensities, and **CARMA+conformal** is CARMA with the best conformal cost view. **Scenario-MPC** is a demand-aware comparator that uses the same information as CARMA in a different way. At every hour it draws K arrival scenarios for the next ℓ hours from the historical job traces from which CARMA’s profile is estimated (the jobs of one historical week, or trace day, at the same hours), and solves a two-stage stochastic programme by sample-average approximation: the first-hour allocation of the pending jobs is shared, while the later allocation and the scenario jobs are planned separately in each scenario, and the mean cost is minimised. Unserved scenario work carries CARMA’s ghost penalty Π_g . The coupling across scenarios breaks the network structure, so Scenario-MPC is solved as an LP with HiGHS; (K, ℓ) are tuned on validation data. **Oracle** solves (2)–(5) with full knowledge of arrivals and intensities.

3.5 Data and experimental design

Carbon intensity

We use half-hourly regional carbon intensity for Great Britain from the Carbon Intensity API of the National Energy System Operator (NESO) [45] from 1 January 2022 to 31 December 2024. Values are averaged to hourly resolution. NESO does not meter regional intensity. For past periods, the regional endpoint returns a single modelled estimate per half-hour (in a field named `forecast`), and we treat this estimate as the realised intensity. The oracle and the “Perfect” variants are therefore clairvoyant with respect to NESO’s published regional values, and all emission figures inherit their modelling error. NESO’s forward regional forecasts are not archived, so they cannot serve as a historical benchmark for our forecaster. Four regions host the candidate sites: South Scotland, North West England, West Midlands and London. They span the north–south gradient of the British grid, from wind- and nuclear-rich South Scotland to gas-dependent London, and include the main data-centre markets of London, Manchester (North West England) and Birmingham (West Midlands). The largest cluster of the London market, around Slough, lies in NESO’s South England region; we use the London region as a stylised proxy for a southern, gas-dependent site. Our first download missed 602 of the 26,304 hours in at least one region, most of them in two blocks of eleven to twelve days at the start of 2023 and 2024. Re-requesting these days in one-day chunks recovered all but 52 hours, which the API does not provide: 41 hours on 20–22 October 2023 (in the tuning period) and 11 hours on 11–12 June 2024. These are filled causally by carrying the last observation forward, so no filled value uses later data. Filled hours are excluded from forecast training, calibration, ACI feedback and forecast evaluation. In the scheduling simulation they act as realised intensities; the two test weeks whose episodes contain them are checked separately in Section 5.3. The period up to 30 June 2023 is used to train the forecaster, July–September 2023 to calibrate the conformal scores, October–December 2023 to tune the scheduler, and all 52 weeks starting on Mondays in 2024 as the out-of-sample test set.

Sites and energy parameters

Batch capacities are 300, 400, 400 and 600 servers for South Scotland, North West England, West Midlands and London. This reflects a small, clean northern site and a large site in London; the split is varied in the sensitivity analysis. Interactive load takes $20\% \pm 15\%$ of each site's capacity on a diurnal cycle that peaks in the afternoon. We assume $e = 0.4$ kWh per server-hour. The assumed PUE values are 1.15, 1.20, 1.25 and 1.30 from north to south, reflecting free-cooling potential. Migrated work moves $\delta = 2$ GB per server-hour at $\eta = 0.02$ kWh/GB. This value lies within the wide range of published estimates of transmission energy intensity [62]. The main experiments count only the energy of busy servers, which assumes that idle servers are switched off or reallocated. The sensitivity analysis varies η and the capacity split, and it adds an always-on case with an idle draw of 0.2 kWh per server-hour for every batch server.

Workloads

Job arrivals follow a non-homogeneous Poisson process with a diurnal peak in the afternoon and 30% lower weekend rates. Jobs originate at the four sites with probabilities 0.10, 0.20, 0.25 and 0.45. Width m_j is log-uniform on $[4, 64]$ servers and the run time at full width r_j is uniform on $\{1, \dots, 6\}$ hours, so $w_j = m_j r_j$. A share v of jobs is urgent (deadline = $a_j + r_j$, no temporal slack); for the others the window is $\lceil f_j r_j \rceil$ hours with $f_j \sim U[1.5, 4]$, capped at 24 hours. The baseline value is $v = 0.3$. The arrival rate is scaled so that the offered load equals a fraction $\rho \in \{0.3, 0.5, 0.7\}$ of the mean free capacity of the fleet. Each test episode covers one week of arrivals. We also simulate 24 hours of arrivals before and after the week, so the evaluated week starts and ends under steady-state congestion. Metrics are computed only for jobs released during the week. For each week and load level we draw three independent workload traces with fixed seeds, and every policy runs on the same traces. The demand profile used by CARMA is estimated from 13 independent historical traces of the same generator. This favours anticipation, because the profile is then an unbiased estimate of the arrival rate. We therefore also evaluate a trace-driven workload whose profile is learned from different days (next paragraph) and profiles learned at the wrong load (Section 5.5). We study two regimes: *spatio-temporal*, where migration is allowed, and *temporal-only*, where it is not. All hyperparameters are tuned separately for each regime.

Trace-derived workload

To test anticipation on real, non-stationary demand, we derive a fluid workload from the batch-task table of the Alibaba 2018 cluster trace [63, 64]. The table has 14.3 million task records over eight days. We keep the 13.3 million terminated tasks with valid timestamps and a planned CPU request and group them by job (one directed acyclic graph of tasks), which gives 3.85 million jobs (script `prepare_trace.py`). The trace records no submission times, so a job's arrival is the start of its first task. Its work is the sum over tasks of instances $\times$ run time $\times$ planned CPU request (core-hours), and its width is the largest planned core count of a single task; planned requests are not measured utilisation. Both are converted to servers of 96 cores and scaled so that the offered load in the evaluation days equals $\rho = 0.5$. Task precedence within a job is ignored. The trace records no deadlines, so windows are drawn as in the synthetic

model: 30% of jobs are urgent, with a window equal to their run time rounded up to whole hours, and the others receive 1–23 hours of slack. Origins are assigned at random with the site probabilities above. To keep the planning problems tractable, jobs are aggregated into batches by arrival hour, origin and window class, and the works and widths of a batch are added. Pooling widths relaxes the per-job width limits, because a batch may run its short, wide members and its long, narrow members at a combined rate that no single member could sustain. The batched workload is therefore an approximation, and we test its effect with finer batches that also separate jobs by run-length class (Section 5.4). CARMA and Scenario-MPC learn from days 1–4 of the trace and are evaluated on days 5–8, preceded by 24 hours of warm-up. Days 5–8 carry 39% more work than days 1–4, so the learned hour-of-day profile underestimates evaluation demand by 28%, and hourly work ranges from 0.2 to 3.6 times its mean, so bursts are not anticipated. The four evaluation days are paired with each of the 52 test weeks of 2024, starting on the Monday, with three independent draws of origins and deadlines. These 156 episodes vary the carbon conditions and the synthetic attributes, but they share one demand chronology.

Metrics and statistics

The primary metric is operational emissions (tCO₂) per week under average regional intensity, which is an *attributorial* measure (Section 6 discusses marginal emissions). We report the reduction relative to ASAP-Local, the gap to the clairvoyant oracle, the share of late work, the share of work executed away from its origin and the decision time. Reductions and gaps are means of per-episode ratios, so they cannot be recomputed from the mean emissions in the tables. Late work is work still outstanding when its deadline passes; online policies keep processing it, all of it is completed before the end of the simulation, and its emissions are counted. The gap to the oracle is an optimality gap only for schedules that meet all deadlines. Where lateness is negligible, as in the spatio-temporal regime, it measures the loss of an online policy against the optimum. Where it is not, emissions and service must be read together, and we also report a service-adjusted gap. It charges each late server-hour a penalty equal to the largest local unit cost of the episode, in addition to its emissions, and compares with an oracle that minimises the same objective: evaluated work may be left undone at that penalty, while boundary work must still be served on time. The on-time part of any schedule that serves its boundary work on time is feasible for this oracle, so its value bounds the adjusted emissions of such schedules from below. The estimand for comparisons is the mean over the 52 test weeks of the paired difference in gap between a policy and CARMA, using week-level means over the three traces. Consecutive weeks are serially correlated (lag-1 autocorrelation of the paired differences ≈ 0.5), so confidence intervals and two-sided p -values come from a moving-block bootstrap over weeks [65] with blocks of four weeks and 5,000 draws. The p -value is the share of bootstrap means of the centred series that are at least as far from zero as the observed mean. We report block lengths of two and eight weeks as a sensitivity check. p -values are Holm-adjusted jointly over all comparisons of the main experiment (both regimes and all loads) and separately within the trace and sensitivity experiments.

599 4 Theoretical guarantees

600 This section establishes a worst-case limit for *every* online scheduler, prediction-
601 dependent guarantees for CARMA, and a fairness property of the carbon attribution
602 induced by the pooled scheduling problem. Proofs are in Appendix A. Throughout,
603 unit costs (Eq. (1), including any backstop resource introduced below) satisfy $0 < L \leq$
604 $c_{os}(t) \leq U$, and $\theta = U/L$ is the *unit-emission-cost ratio*. It refers to unit costs, which
605 include energy use, PUE, transfer overheads and any backstop, not to grid intensities
606 alone. Operational costs alone can be zero, because the hourly regional intensity of
607 South Scotland is 0 gCO₂/kWh in 133 hours of our 2022–2024 data. A positive floor
608 appears once the embodied emissions of the servers are allocated per server-hour,
609 which is an accounting convention rather than a claim that scheduling avoids embodied
610 emissions. As an illustration, a floor of 30 gCO₂e added to every unit cost, with the
611 largest operational unit cost observed in 2024 (221 g), gives $\theta = 251/30 \approx 8.4$. An
612 online algorithm is Γ -*competitive* if its emissions never exceed Γ times those of the
613 clairvoyant oracle, for any arrival sequence and any intensity trajectory.

615 4.1 A universal lower bound under demand uncertainty

616 Competitive analyses of carbon-aware scheduling model uncertainty about future
617 *prices*, that is, intensities, for a single workload. Examples are online search and
618 one-way trading [24], the online pause-and-resume problem [25], and spatiotemporal
619 allocation with deadlines (CarbonClipper) [26]. When several jobs share scarce low-
620 carbon capacity, difficulty remains even when all prices are known in advance, because
621 future *demand* is unknown.

622 **Theorem 1** (Lower bound) *Let $\theta > 1$. No online algorithm (deterministic or randomised,*
623 *with integral or fractional decisions) can be better than $R^*(\theta)$ -competitive, even if it knows all*
624 *future carbon intensities exactly, where*

$$625 R^*(\theta) = \max_{1 \leq p \leq \theta} \left\{ 1 + \frac{(p-1)(\theta-p)}{p^2 - p - 1 + \theta} \right\} \geq 1 + \frac{\sqrt{\theta}(\sqrt{\theta}-1)^2}{2\theta - \sqrt{\theta} - 1} = \Omega(\sqrt{\theta}). \quad (8)$$

626 *On the same family of instances, the competitive ratio of myopic receding-horizon scheduling*
627 *(MPC) is at least $(1 + \theta)/2$, a value approached as $p \rightarrow 1$.*

628 For $\theta = 1$ every schedule costs the same and $R^*(1) = 1$; both expressions in (8)
629 tend to 1 as $\theta \downarrow 1$.

630 The bound is established on a two-hour instance and is a quantitative version of
631 Proposition 2, which treats the case $L = 0$. Any algorithm must decide how much
632 flexible work to run now at moderate cost before it learns whether an urgent job will
633 claim the cheap capacity later. The adversary thus raises the effective price of the clean
634 hour from L to U . The resulting order, $\sqrt{\theta}$, is the same as for deterministic online
635 min-search with unknown prices [24]. The difference is where it comes from: here every
636 price is known, and the loss is caused only by the unknown arrival of a competing job.
637 For the illustrative $\theta \approx 8.4$ above, $R^* = 1.81$, whereas myopic MPC can be driven
638 to 4.7. For $\theta = 100$ the values are 5.29 and 50.5. $R^*(\theta)$ is a lower bound derived

from one family of instances; we do not know whether any online algorithm attains it. Theorem 1 is a worst-case statement, but it points in the same direction as the average-case evidence of Section 5.3. Carbon forecasts cannot substitute for information about demand, and this motivates the learning-augmented design of CARMA [27].

4.2 Deadline safety, consistency and smoothness of CARMA

We analyse CARMA under two assumptions.

- (A1) *Backstop capacity.* In every hour, work may also run on a backstop resource of unlimited capacity at a unit cost in $[L, U]$, for example by bursting to a public-cloud region or to reserve servers at a carbon-intensive site. The backstop's unit cost is known in advance, or its forecast is clipped to $[L, U]$, so the planner never prices it above U . All jobs, real and predicted, are *window-feasible*, $w_j \leq m_j(d_j - a_j)$, and have windows of at most H hours, so every planning problem contains all pending deadlines (this holds by construction in our workloads). The penalties satisfy $\Pi > U$ for real jobs and $\Pi_g \geq U$ for ghost jobs. The implementation enforces window-feasible ghost jobs (Section 3.3) and sets Π_g to 1.5 times the largest local unit cost, which exceeds every unit cost including transfer overheads.
- (A2) *Episode within the planning window.* The episode $\{0, \dots, T-1\}$ has length $T \leq H$, as for a daily batch window, so every planning problem extends to the end of the episode.

To measure prediction quality at hour t , let F_t be the set of jobs that actually arrive after t , and $\hat{F}_t$ the ghost jobs used by CARMA. Jobs are compared after splitting them into *atoms* with proportional widths, which leaves every planning problem unchanged (Appendix A). Let d_t be the smallest total work of atoms that must be added to or removed from $\hat{F}_t$ to obtain F_t . Let $\varepsilon_t = \max_{\tau > t, o, s} |\hat{c}_{os}(\tau | t) - c_{os}(\tau)|$ be the carbon-cost forecast error, and W_t the total work pending at t plus the larger of the true and predicted future work.

Proposition 3 (Baseline robustness) *Under (A1), every policy that completes all work by its deadlines is θ -competitive.*

Theorem 2 (Guarantees for CARMA) *Under (A1), for any arrival sequence and any predictions, CARMA completes every job by its deadline (deadline safety) and is therefore θ -competitive. If (A2) also holds, then*

$$\text{ALG} \leq \min \left\{ \theta \cdot \text{OPT}, \text{OPT} + \sum_{t=0}^{T-1} [(U - L) d_t + 2 \varepsilon_t W_t] \right\}. \quad (9)$$

In particular, CARMA is optimal with exact predictions ($d_t = \varepsilon_t = 0$), which is consistency, and its excess emissions grow at most linearly in the prediction errors, which is smoothness.

These properties should be read with care. Robustness by itself does not distinguish CARMA: by Proposition 3 it holds for any policy that meets its deadlines. What CARMA adds is deadline safety under *arbitrary* ghost predictions, together with

consistency and smoothness. Deadline safety holds because ghost jobs carry a softer penalty than real jobs and the backstop can always absorb real work, so a misprediction can divert real work to dirtier capacity but never make it late. Without a backstop this is no longer true. With one server and two hours, a flexible job that could run in the first hour may be deferred to a cleaner second hour, and an unexpected urgent job in the second hour then makes one of them late, although an offline schedule meets both deadlines. A soft ghost penalty does not create capacity. “Exact predictions” in the theorem means exact predictions of both demand and carbon; the CARMA-Perfect variant of the experiments has perfect carbon forecasts but still uses the average demand profile and a finite look-ahead, so it is not the case $d_t = 0$. The smoothness term charges demand errors at the spread $U - L$ of unit costs, not at the lateness penalty. The term is informative when predictions are close to the realised demand. For random arrivals, however, an expected-value profile never matches the realised jobs exactly. Moreover, each misprediction enters d_t in every epoch in which it is visible, so the bound in (9) can be loose. In the main experiments, assumption (A2) is not satisfied (weekly episodes with $H = 24$ h), no backstop is modelled, and some hours have zero operational cost. The theorem therefore describes an idealised setting and does not certify the reported numbers; the degradation profile in the weekly experiments is tested empirically in Section 5.6. Appendix C checks the theorem numerically in one-day episodes that do satisfy (A1) and (A2). Robustness θ is also far from the $\Omega(\sqrt{\theta})$ lower bound. Closing this gap while retaining consistency is left for future work, for example by protecting deferral decisions with reservation prices [24, 26]. Fig. 2 summarises the worst-case landscape.

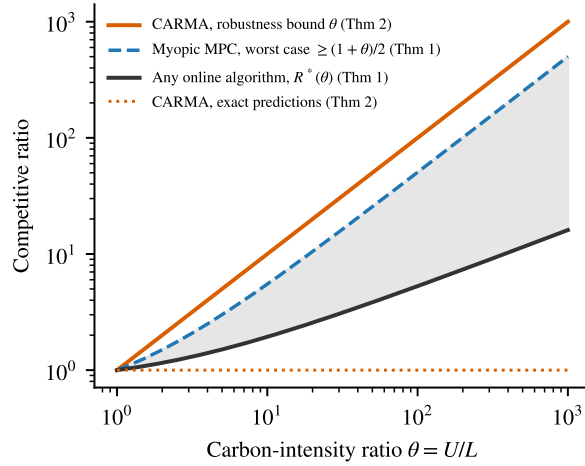


Fig. 2 Worst-case landscape as a function of the unit-emission-cost ratio $\theta = U/L$. The curves refer to different information regimes. Without information about future demand, no online algorithm can lie below $R^*(\theta)$, even with exact carbon forecasts (Theorem 1), and myopic MPC can be driven to $(1 + \theta)/2$. Given a backstop (A1), every policy that meets its deadlines, CARMA included, is within θ (Proposition 3). With exact predictions of *both* demand and carbon and an episode within the planning window (A1–A2), CARMA is optimal (Theorem 2).

4.3 Fair carbon attribution among sites

Pooling capacity reduces total emissions but redistributes them. Work submitted in London is executed in South Scotland, and Scottish jobs may be displaced by it. The organisations that operate the sites, or the business units that submit the jobs, therefore need an attribution of the pooled emissions that each of them can accept. This is an *internal* allocation problem, like carbon transfer pricing between units. It is distinct from external GHG Protocol reporting, where Scope 2 emissions are reported by the entity that consumes the electricity [66].

We model the allocation as a cooperative game. The players are the sites $\mathcal{S}$. A coalition $C \subseteq \mathcal{S}$ pools the capacity of its sites and serves the jobs submitted at them. Its stand-alone value $c(C)$ is the optimum of (2)–(5) restricted to those jobs and sites, which includes the lateness penalty if the coalition cannot serve its own jobs on time. An attribution $\phi \in \mathbb{R}^{|\mathcal{S}|}$ is *budget-balanced* if $\sum_o \phi_o = c(\mathcal{S})$. It is in the *core* if $\sum_{o \in C} \phi_o \leq c(C)$ for every coalition C . A core attribution gives no group of sites a reason to leave the pool and schedule on its own.

Let (λ, ν, μ) be optimal dual variables of the grand-coalition problem: λ_j for the work constraint (3), $\nu_{jt} \leq 0$ for the width constraint (4) and $\mu_{st} \leq 0$ for the capacity constraint (5). The *dual carbon attribution* is

$$\phi_o = \sum_{j: o_j=o} \left(\lambda_j w_j + \sum_t \nu_{jt} m_j \right) + \sum_t \mu_{ot} K_o(t). \quad (10)$$

Site o is charged the carbon shadow price of the work it submits and is credited ($\mu \leq 0$) with the shadow value of the capacity it contributes. A site whose clean capacity is worth more than the emissions of its own jobs therefore receives a net credit, that is, a negative internal charge.

Theorem 3 (Core-stable carbon attribution) *The attribution (10) is budget-balanced and in the core of the pooling game. In particular, it is individually rational: no site is charged more than its stand-alone optimum $c(\{o\})$.*

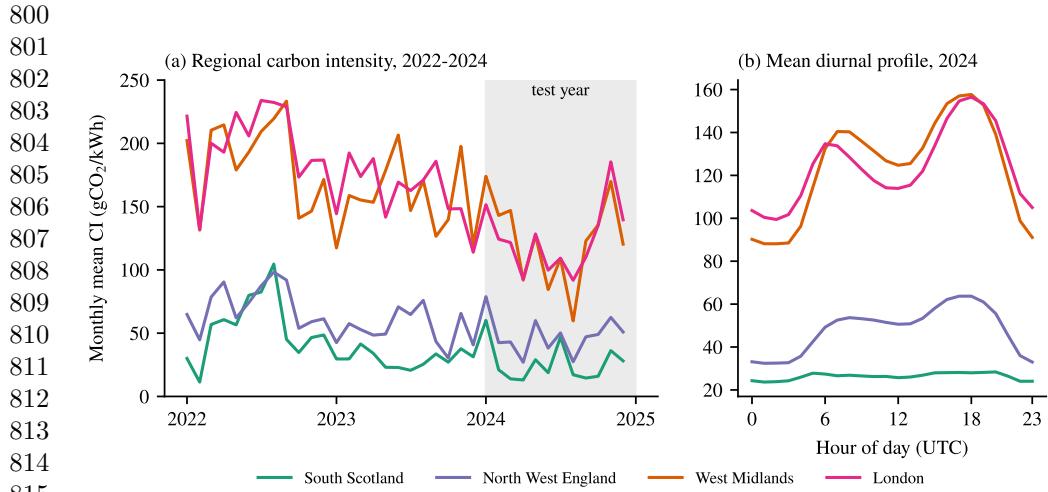
The result follows from Owen’s theorem on linear production games [31]; games defined by network flows are known to be totally balanced [32]. It concerns the penalised hindsight game defined above. Core stability is one requirement of an acceptable allocation; it does not imply uniqueness, incentive compatibility or other notions of fairness, and it does not extend automatically to the realised emissions of an online policy. The duals are those of the hindsight problem, which suits after-the-fact accounting. They are available as the optimal node potentials of the min-cost flow; in our experiments we obtain them from an LP solver. Optimal duals need not be unique, so Section 5.7 reports the range of (10) over all optimal duals. It also compares the dual attribution with the Shapley value [67], which is budget-balanced but not guaranteed to lie in the core, and with simpler rules. The simpler rules are *origin-based* (each site is charged the emissions of its own jobs wherever they ran), *host-based* (each

783 site is charged the emissions of the work it executes, the analogue of location-based
784 Scope 2 reporting), and proportional rules.

786 5 Results

788 5.1 Carbon-intensity data and forecasts

789 Fig. 3 shows the regional intensities. Over 2022–2024, mean intensity was 37 gCO₂/kWh
790 in South Scotland, 58 in North West England, 156 in the West Midlands and 163
791 in London. The regional means indicate large spatial heterogeneity, and it persists
792 hour by hour: the spread between the cleanest and the dirtiest of the four sites has
793 a median of 156 gCO₂/kWh and exceeds 54 g in 90% of hours. This is what makes
794 spatial shifting attractive. Annual mean intensities declined at every site; London’s fell
795 from 202 gCO₂/kWh in 2022 to 125 in 2024. Diurnal profiles peak in the early evening
796 (around 18:00 UTC), and the profile of South Scotland is much flatter than those of the
797 southern sites. The four series are positively correlated (pairwise correlations between
798 0.51 and 0.87), because national wind output drives all of them.



816 **Fig. 3** Hourly carbon intensity of the four candidate sites (NESO regional data). (a) Monthly means,
817 2022–2024; the shaded area is the out-of-sample test year. (b) Mean diurnal profile in 2024 (UTC).

819 The forecaster was trained on 1.24 million (issue time, site, horizon) samples in
820 about one minute. In 2024 its mean absolute error (MAE), averaged over horizons
821 and sites, is 31.7 gCO₂/kWh, against 35.3 for persistence and 40.9 for a seasonal-
822 naive forecast. The advantage is 23–25% at horizons of 1–6 hours, and the difference
823 is significant at all English sites up to 18 hours in Diebold–Mariano tests [68]. Day-
824 ahead skill is limited, because no weather inputs are used. Adaptive recalibration
825 keeps the annual coverage of the upper bounds within 0.003 of target despite the 2024
826 decarbonisation drift. Appendix B gives the full evaluation (accuracy by horizon and
827 site, coverage, pinball loss).

5.2 Hyperparameter tuning

All policies were tuned on 13 validation weeks (October–December 2023, $\rho \in \{0.3, 0.5, 0.7\}$), separately for each regime. Fig. 4 shows the spatio-temporal regime. Anticipating demand helps across the whole grid: every tested configuration beats myopic MPC, whose mean gap to the oracle is 22.3%. The gap is smallest around $\kappa = 1$, where expected demand is anticipated without inflation: 9.0% for $\ell = 12$ and $\ell = 18$. Over-anticipation ($\kappa = 1.25$ with $\ell = 18$) reserves too much clean capacity and is penalised (12.1%). We selected $\kappa = 1$ and $\ell = 12$ hours, resolving configurations within 0.25 percentage points of the best towards the shorter look-ahead. In the temporal-only regime, anticipation brings no benefit. The best configuration ($\kappa = 0.5$, $\ell = 6$; 4.50%) is indistinguishable from MPC (4.50%), and larger κ is slightly worse. For MPC-Protect, the best protection level was $r = 0.3$ with $\tau = 1$ h in the spatio-temporal regime (21.0% against 22.3% for MPC). In the temporal-only regime every protection level was worse than MPC (at best 6.0%).

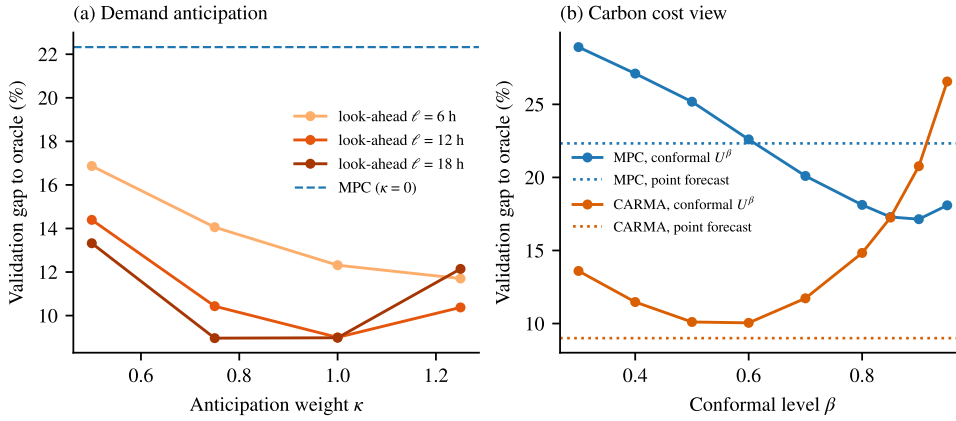


Fig. 4 Validation results for the spatio-temporal regime (2023-Q4, mean over 13 weeks and three load levels). (a) Gap to the oracle as a function of the anticipation weight κ and look-ahead ℓ , using point forecasts. (b) Effect of the carbon cost view: conformal quantiles U^β against point forecasts, for myopic MPC and for CARMA ($\kappa = 1$, $\ell = 12$).

The cost view interacts with anticipation (Fig. 4b). For myopic MPC, high conformal levels help. The gap falls from 25.2% at $\beta = 0.5$ to 17.1% at $\beta = 0.9$ and rises again at $\beta = 0.95$ (18.1%), against 22.3% with point forecasts. For CARMA, every conformal level is worse than the point forecast: the best, $\beta = 0.6$, gives 10.0% against 9.0%. The mechanism is not a uniform risk premium. On the validation weeks, the $\beta = 0.6$ view lies on average only 0.8 g below to 4.9 g above the point forecast at South Scotland, North West England and London, but 17.8 g above it at the volatile West Midlands site. The offset also grows with the horizon, and at $\beta = 0.8$ it rises to 28–128 g at a 24-hour horizon. High levels therefore make distant hours and volatile sites look dirtier, so the planner runs work now and at stable sites, which keeps future clean capacity free. For a myopic planner this is a crude form of anticipation. Once CARMA represents

future demand explicitly, the same distortion double-counts it and gives up useful deferrals. The selected CARMA therefore uses point forecasts. For the test we keep MPC+conformal at $\beta = 0.9$ and CARMA+conformal at $\beta = 0.6$ (spatio-temporal) as ablations. In the temporal-only regime, $\beta = 0.5$ was the best conformal view for both.

5.3 Out-of-sample scheduling performance

Tables 2 and 3 report results for the 52 test weeks of 2024, with three independent workload traces per week. Weekly traces contain on average 917, 1,527 and 2,141 jobs (69,400, 115,400 and 162,000 server-hours) at $\rho = 0.3, 0.5$ and 0.7 . Fig. 5 shows the gaps to the oracle.

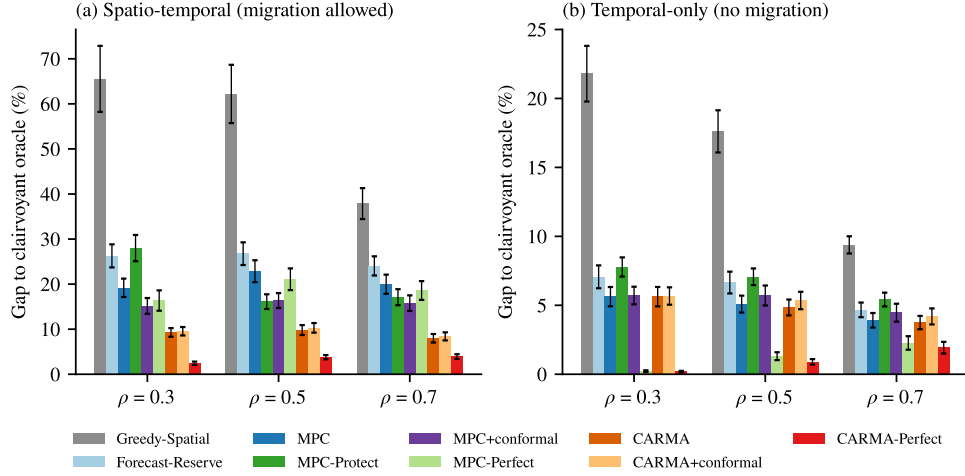


Fig. 5 Mean gap to the clairvoyant oracle over the 52 test weeks of 2024 (week means of three traces) with 95% confidence intervals. ASAP-Local is omitted (gap 72–280% with migration, 9–22% without).

Spatio-temporal regime

With migration allowed, the oracle cuts attributional emissions relative to ASAP-Local by 70.8%, 57.5% and 40.6% at $\rho = 0.3, 0.5$ and 0.7 . Greedy-Spatial realises only part of this potential (gaps of 38–66%), because it has no temporal flexibility and floods the cleanest site with whatever arrives first. The forecast-driven planners come closer. Forecast-Reserve has gaps of 24–27%, MPC 19–23%, and the two stronger baselines, MPC-Protect and MPC+conformal, 15–28%. MPC-Protect is the best baseline at $\rho = 0.5$ (16.1%) and MPC+conformal at the other loads (15.2% and 15.8%). CARMA reduces emissions by 68.3%, 53.6% and 36.1% and has gaps of 9.3%, 9.8% and 8.0%. Relative to MPC this is a further emission reduction of 7.4%, 9.3% and 8.8%. Its advantage over the best baseline at each load is 5.9 (95% block-bootstrap CI 4.3–8.2), 6.3 (5.2–7.3) and 7.8 (6.7–8.9) percentage points of gap. CARMA emits less than MPC in 51, 51 and 52 of the 52 week-means. All comparisons with online baselines are

significant after Holm correction ($p < 0.001$). CARMA+conformal is 0.3–0.5 points worse than CARMA ($p \leq 0.04$).

CARMA also finishes more work on time under heavy load. At $\rho = 0.7$, 0.03% of its work is late, against 1.9% for MPC, 0.7% for MPC-Protect, 1.1% for Forecast-Reserve and 4.5% for ASAP-Local. The service-adjusted gap therefore widens the difference: 8.1% for CARMA against 17.9–30.9% for the forecast-driven baselines. CARMA migrates about the same share of work as MPC (31–70%), whereas MPC-Protect migrates more (38–81%). CARMA’s advantage thus comes from better timing of the use of scarce clean capacity, not from moving more data. For the fleet of 1,700 batch servers at $\rho = 0.5$, CARMA emits 0.30 tCO₂ per week less than MPC, about 16 tCO₂ per year, and 3.16 tCO₂ per week less than carbon-agnostic operation. These absolute figures scale with fleet size.

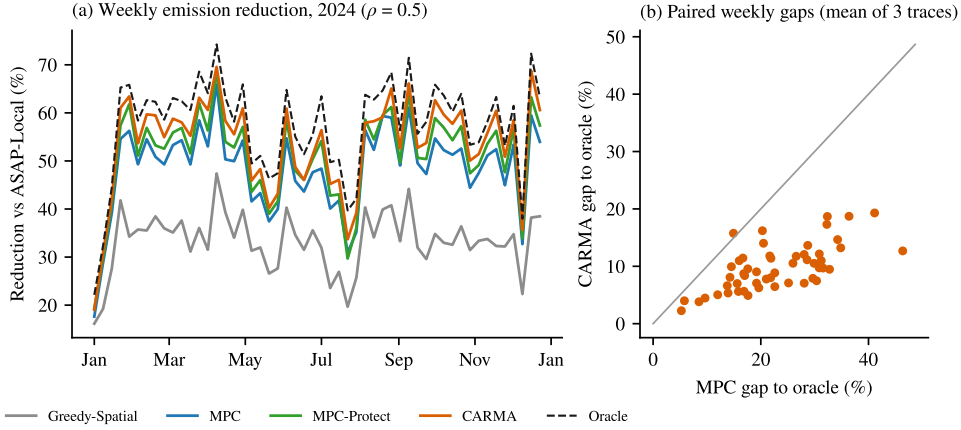


Fig. 6 Weekly results in 2024 (spatio-temporal regime, $\rho = 0.5$, means of three traces). (a) Emission reduction relative to ASAP-Local. (b) Paired weekly gaps to the oracle; points below the diagonal are weeks in which CARMA outperforms MPC.

Value of information

The clairvoyant variants separate two sources of loss. With perfect intensity forecasts, the myopic MPC-Perfect still has gaps of 16.4%, 21.1% and 18.6%. Perfect carbon foresight is thus worth only 1.4–2.8 percentage points to a myopic planner, whereas anticipating demand is worth 9.9–13.1 points (MPC versus CARMA). With both, CARMA-Perfect comes within 2.4–3.9% of the oracle. The two kinds of information are complementary. Once demand is anticipated, forecast accuracy is worth 4.1–6.8 points (CARMA versus CARMA-Perfect), about two to four times more than for the myopic planner. The ranking is stable across seasons (Fig. 6a), although the level of achievable reductions varies from week to week with wind conditions.

967 *Temporal-only regime*

968 Without migration (Table 3), each job can only move in time at its origin. Achievable
969 reductions are much smaller: the oracle saves 17.6%, 14.8% and 8.5%. MPC and
970 CARMA are within 3.7–5.6% of the oracle. They are statistically indistinguishable at
971 $\rho = 0.3$ (difference 0.00 points, CI -0.02 to 0.02). CARMA is slightly better at higher
972 loads, by 0.24 (0.15–0.37) and 0.16 (0.08–0.27) points. Both are better than Forecast-
973 Reserve and MPC-Protect. Anticipation brings little benefit here, because each site’s
974 capacity serves only its own jobs, and flexible and urgent work rarely compete for
975 the same clean hours in a way that deferral cannot absorb. The remaining loss is due
976 almost entirely to forecast error: the Perfect variants close the gap to 0.2–2.3%. Heavy
977 load exposes a trade-off that the emission metric hides. At $\rho = 0.7$, the capacity at
978 the origin sites is itself insufficient, and every carbon-aware policy delays work: 9.2%
979 of CARMA’s work and 12.0% of MPC’s is late, against 4.5% for ASAP-Local. Under
980 the service-adjusted metric, carbon-agnostic operation (22.3%) is then preferable to
981 MPC (42.6%) and CARMA (32.6%). Without migration and near saturation, temporal
982 shifting should therefore be restricted or combined with explicit service constraints.
983

984 *Computation*

985 A CARMA decision takes 9–39 ms on one CPU core, against 2–6 ms for MPC; the
986 difference comes from the ghost jobs, which enlarge the planning network. This is
987 negligible for hourly decisions. A clairvoyant weekly oracle with up to about 2,800 jobs
988 is solved exactly in under one second by network simplex.
989

990
991
992
993
994
995
996
997
998
999
1000
1001
1002
1003
1004
1005
1006
1007
1008
1009
1010
1011
1012

Table 2 Out-of-sample results in the spatio-temporal regime (migration allowed), 52 weeks of 2024×3 traces. Emis.: mean weekly operational emissions (tCO₂, attributional). Red.: reduction relative to ASAP-Local (%). Gap: excess over the clairvoyant oracle (%). Adj.: service-adjusted gap, in which late work is additionally charged at the week’s largest local unit cost. Late: share of work finished after its deadline (%). Mig.: share of work executed away from its origin (%). p : two-sided Wilcoxon signed-rank test of week-mean emissions against CARMA, Holm-adjusted over all comparisons in Tables 2 and 3 of the same regime. The best gap among implementable online policies is in bold.

Load	Policy	Emis.	Red.	Gap	Adj.	Late	Mig.	p
$\rho = 0.3$	ASAP-Local	3.70	0.0	280.1	280.1	0.00	0.0	<0.001
	Greedy-Spatial	1.75	54.1	65.5	65.5	0.00	81.7	<0.001
	Forecast-Reserve	1.38	63.9	26.3	26.3	0.00	83.6	<0.001
	MPC	1.31	65.8	19.2	19.2	0.00	69.4	<0.001
	MPC-Protect	1.40	63.4	28.0	28.0	0.00	80.5	<0.001
	MPC+conformal	1.27	66.8	15.2	15.2	0.00	70.0	<0.001
	MPC-Perfect	1.28	66.6	16.4	16.4	0.00	69.1	<0.001
	CARMA	1.22	68.3	9.3	9.3	0.00	70.1	–
	CARMA+conformal	1.22	68.3	9.5	9.5	0.00	70.4	0.037
	CARMA-Perfect	1.15	70.2	2.4	2.4	0.00	69.9	<0.001
	Oracle	1.13	70.8	0.0	0.0	0.00	70.9	–
$\rho = 0.5$	ASAP-Local	6.08	0.0	147.7	148.2	0.05	0.0	<0.001
	Greedy-Spatial	4.12	33.3	62.2	62.2	0.00	76.4	<0.001
	Forecast-Reserve	3.32	46.9	26.8	26.9	0.02	78.2	<0.001
	MPC	3.22	48.5	22.9	23.0	0.01	50.0	<0.001
	MPC-Protect	3.07	51.1	16.1	16.1	0.00	58.7	<0.001
	MPC+conformal	3.08	51.1	16.4	16.4	0.00	50.6	<0.001
	MPC-Perfect	3.18	49.3	21.1	21.2	0.01	49.4	<0.001
	CARMA	2.92	53.6	9.8	9.8	0.00	50.5	–
	CARMA+conformal	2.94	53.4	10.3	10.3	0.00	50.6	<0.001
	CARMA-Perfect	2.79	56.0	3.8	3.8	0.00	50.2	<0.001
	Oracle	2.70	57.5	0.0	0.0	0.00	51.9	–
$\rho = 0.7$	ASAP-Local	8.29	0.0	71.7	98.5	4.54	0.0	<0.001
	Greedy-Spatial	6.76	19.1	37.9	37.9	0.00	73.3	<0.001
	Forecast-Reserve	6.14	26.9	24.0	30.6	1.12	73.5	<0.001
	MPC	5.94	29.3	20.0	30.9	1.89	31.6	<0.001
	MPC-Protect	5.82	30.9	17.1	21.0	0.69	37.8	<0.001
	MPC+conformal	5.76	31.7	15.8	17.9	0.34	32.1	<0.001
	MPC-Perfect	5.87	30.1	18.6	30.1	1.93	31.4	<0.001
	CARMA	5.41	36.1	8.0	8.1	0.02	31.5	–
	CARMA+conformal	5.44	35.8	8.4	8.5	0.01	31.6	<0.001
	CARMA-Perfect	5.24	38.4	3.9	4.2	0.04	31.0	<0.001
	Oracle	5.07	40.6	0.0	0.0	0.00	32.8	–

1059
1060
1061
1062
1063
1064
1065
1066
1067
1068
1069
1070
1071
1072
1073
1074
1075
1076
1077
1078
1079
1080
1081
1082
1083
1084
1085
1086
1087
1088
1089
1090
1091
1092
1093
1094
1095
1096
1097
1098
1099
1100
1101
1102
1103
1104

Table 3 Out-of-sample results in the temporal-only regime (no migration), 52 weeks of 2024×3 traces. Columns as in Table 2. Greedy-Spatial is omitted because without migration it coincides with ASAP-Local.

Load	Policy	Emis.	Red.	Gap	Adj.	Late	p
$\rho = 0.3$	ASAP-Local	3.70	0.0	21.8	21.8	0.00	<0.001
	Forecast-Reserve	3.27	11.8	7.1	8.6	0.38	<0.001
	MPC	3.23	13.0	5.6	5.6	0.00	0.743
	MPC-Protect	3.30	11.3	7.8	7.8	0.00	<0.001
	MPC+conformal	3.24	12.9	5.7	5.8	0.01	0.232
	MPC-Perfect	3.09	17.4	0.2	0.3	0.01	<0.001
	CARMA	3.23	13.0	5.6	5.6	0.00	–
	CARMA+conformal	3.24	13.0	5.7	5.7	0.01	0.232
	CARMA-Perfect	3.09	17.4	0.2	0.2	0.01	<0.001
	Oracle	3.08	17.6	0.0	0.0	0.00	–
$\rho = 0.5$	ASAP-Local	6.08	0.0	17.6	17.8	0.05	<0.001
	Forecast-Reserve	5.54	9.2	6.6	18.3	3.06	<0.001
	MPC	5.47	10.5	5.1	6.9	0.53	<0.001
	MPC-Protect	5.57	8.8	7.1	8.8	0.50	<0.001
	MPC+conformal	5.50	9.9	5.7	8.4	0.75	<0.001
	MPC-Perfect	5.30	13.7	1.3	4.1	0.76	<0.001
	CARMA	5.46	10.7	4.8	6.1	0.38	–
	CARMA+conformal	5.48	10.2	5.3	7.3	0.54	<0.001
	CARMA-Perfect	5.28	14.0	0.9	2.9	0.54	<0.001
	Oracle	5.24	14.8	0.0	0.0	0.01	–
$\rho = 0.7$	ASAP-Local	8.29	0.0	9.4	22.3	4.54	<0.001
	Forecast-Reserve	7.94	4.3	4.7	30.7	8.15	<0.001
	MPC	7.88	5.0	3.9	42.6	11.95	0.001
	MPC-Protect	8.00	3.6	5.4	43.0	11.62	<0.001
	MPC+conformal	7.92	4.5	4.5	49.6	13.53	<0.001
	MPC-Perfect	7.77	6.5	2.3	49.4	13.83	<0.001
	CARMA	7.87	5.1	3.7	32.6	9.20	–
	CARMA+conformal	7.90	4.7	4.2	39.8	10.87	<0.001
	CARMA-Perfect	7.75	6.8	1.9	41.2	11.63	<0.001
	Oracle	7.62	8.5	0.0	0.0	0.75	–

5.4 Trace-driven workload

Table 4 reports the Alibaba-trace experiment. Arrivals and job sizes are taken from the trace, and CARMA’s profile is learned from different days, so the profile is biased by 39% and cannot anticipate bursts. The evaluation covers four trace days paired with 52 carbon windows of 2024, with three draws of origins and deadlines each, and 3,072 job batches (65,300 server-hours) per window. Real, bursty demand makes all policies worse than in the synthetic experiment, but the ranking is unchanged. CARMA has a gap of 20.7%. MPC has 36.1%, and MPC-Protect, the best baseline here, 26.7%. Carbon foresight again matters less than demand anticipation: MPC-Perfect reaches only 33.3%, and CARMA’s advantage over MPC-Protect is 6.0 percentage points (95% CI 4.4–8.1; CARMA better in 42 of 52 weeks, $p < 0.001$). The bias of the learned profile costs 2.7 points (CI 1.1–3.8): CARMA with a profile estimated in-sample from the evaluation days reaches 18.0%. With the biased profile but perfect carbon forecasts, the gap would fall by 3.7 points (CARMA-Perfect, 17.0%). The remaining distance to the oracle reflects bursts that no profile anticipates. CARMA also keeps late work low (0.10%, against 0.65% for MPC).

Table 4 Trace-driven workload (Alibaba 2018 cluster trace, days 5–8; profile learned from days 1–4), paired with 52 carbon windows of 2024 $\times$ 3 draws of origins and deadlines; spatio-temporal regime, $\rho = 0.5$. Emissions are per four-day window. Columns as in Table 2; the service-adjusted gap is omitted.

Policy	Emis.	Red.	Gap	Late	Mig.	p
ASAP-Local	3.66	0.0	195.3	0.00	0.0	<0.001
Greedy-Spatial	2.50	32.9	92.5	0.00	75.0	<0.001
Forecast-Reserve	1.94	48.6	42.5	0.81	77.4	<0.001
MPC	1.86	50.9	36.1	0.65	52.8	<0.001
MPC-Protect	1.76	53.8	26.7	0.07	64.0	<0.001
MPC-Perfect	1.82	51.9	33.3	0.63	52.3	<0.001
CARMA	1.69	55.7	20.7	0.10	53.2	–
CARMA-Perfect	1.64	57.2	17.0	0.15	52.2	<0.001
Oracle	1.44	62.8	0.0	0.00	56.7	–
CARMA (in-sample profile)	1.65	56.6	18.0	0.01	52.2	<0.001

5.5 Sensitivity analysis

Table 5 varies one factor at a time around the base case: spatio-temporal regime, $\rho = 0.5$, $v = 0.3$, $\eta = 0.02$ kWh/GB, capacities 300/400/400/600. Each setting uses 26 alternating test weeks of 2024 with two fresh workload traces per week. All policies keep their base-case hyperparameters.

Workload and network

CARMA has the smallest gap of all online policies in every setting. Except under marginal accounting (discussed below), it emits less than MPC in all 26 weeks of each setting. The benefit of anticipation grows with the share of urgent work, because urgent arrivals are precisely the demand that myopic planners crowd out. With $v = 0.5$,

Table 5 Sensitivity analysis (spatio-temporal regime; 26 alternating weeks of 2024×2 traces per setting). Oracle red.: emission reduction of the oracle relative to ASAP-Local (%). Gap columns: excess over the oracle (%). Wins: weeks in which CARMA emits less than MPC. Late: share of late work (%), MPC / CARMA. ^aBusy servers draw 0.2 kWh above an idle draw of 0.2 kWh per server-hour that every batch server incurs regardless of the schedule. ^bSchedules are made with average intensities, but emissions are accounted with the stylised marginal emission factor of Section 5.5, and the oracle optimises marginal emissions.

Setting	Oracle red.	Gap to oracle (%)				Wins vs		Late (%)	
		Greedy-Sp.	F.-Reserve	MPC	MPC-Prot.	CARMA	MPC	MPC	CARMA
Urgent share $v = 0.1$	60.8	77.5	28.3	23.0	16.0	11.9	26/26	0.00 / 0.00	0.00 / 0.00
Urgent share $v = 0.5$	55.4	52.7	25.4	21.5	14.9	6.7	26/26	0.09 / 0.00	0.01 / 0.00
Network energy $\eta = 0.005$ kWh/GB	61.2	68.7	28.5	27.0	17.5	10.9	26/26	0.01 / 0.00	0.01 / 0.00
Network energy $\eta = 0.06$ kWh/GB	52.7	64.1	28.8	19.3	14.5	8.4	26/26	0.01 / 0.00	0.00 / 0.00
Load $\rho = 0.4$, profile learned at 0.5	66.0	74.4	28.9	23.2	15.3	12.1	26/26	0.00 / 0.00	0.00 / 0.00
Load $\rho = 0.6$, profile learned at 0.5	50.3	54.8	27.5	23.8	18.7	8.7	26/26	0.31 / 0.01	0.02 / 0.00
Clean-site capacity halved (150)	51.6	57.1	25.2	21.4	16.6	9.5	26/26	0.01 / 0.00	0.01 / 0.00
Clean-site capacity doubled (600)	67.7	83.0	32.4	28.4	16.9	9.2	26/26	0.00 / 0.00	0.00 / 0.00
London capacity doubled (1200)	46.7	52.0	27.5	22.3	18.5	9.5	26/26	0.01 / 0.00	0.01 / 0.00
Always-on servers (idle 0.2 kWh) ^a	18.2	10.7	4.7	3.5	2.5	1.5	26/26	0.01 / 0.00	0.01 / 0.00
Marginal-emission accounting ^b	13.5	12.0	10.7	8.5	9.1	7.9	17/26	0.01 / 0.00	0.01 / 0.00

CARMA’s gap is 6.7% against 14.9% for MPC-Protect and 21.5% for MPC. With $v = 0.1$, it is 11.9% against 16.0% and 23.0%. The ranking is unchanged over a twelve-fold range of transmission energy intensity ($\eta = 0.005\text{--}0.06$ kWh/GB), and the migrated share of CARMA’s work stays at 49–51%.

Capacity layout

Arrival rates are rescaled with the fleet’s capacity, so the offered load stays at $\rho = 0.5$ of the new capacity. The advantage does not depend on the chosen layout. When the clean South Scotland site is halved (150 servers) or doubled (600), or London’s capacity is doubled, CARMA’s gap stays at 9.2–9.5%. The gaps of MPC (21–28%) and MPC-Protect (16.6–18.5%) stay about two to three times larger. More clean capacity raises the oracle’s reduction (from 51.6% to 67.7%), but a myopic planner still misallocates it.

Misspecified demand profile

CARMA’s ghost jobs come from a profile learned at $\rho = 0.5$. When the actual load is 20% higher ($\rho = 0.6$), so that CARMA under-anticipates, its gap is 8.7% against 18.7% for MPC-Protect and 23.8% for MPC. When the actual load is 20% lower ($\rho = 0.4$), CARMA over-anticipates and reserves more clean capacity than needed. Its gap rises to 12.1%, but it still beats MPC-Protect by 3.2 points (15.3%) and MPC by 11.1 points. The trace experiment of Section 5.4 adds a real, larger (39%) bias. Because ghost work carries a lower penalty than real work, the planner never plans real work late in order to serve ghost demand. Without a backstop, however, a misprediction can still delay real work once capacity is exhausted (Section 4.2); the late-work column of Table 5 tracks this.

Always-on servers

If every batch server draws idle power whether or not it is busy, a large part of the emissions no longer depends on the schedule. The achievable reduction falls to 18.2% for the oracle and 16.9% for CARMA. The ranking is unchanged (gaps of 1.5% for CARMA, 2.5% for MPC-Protect and 3.5% for MPC). Carbon-aware scheduling is therefore most valuable when it is combined with right-sizing, so that deferred work actually reduces the number of powered servers [18].

Marginal emissions

All sites are in one GB market, where gas-fired plant is usually the marginal generator [7, 46]. Moving load between regions then mainly changes *which* region is assigned the average mix, and it abates emissions physically only when low-carbon output would otherwise be curtailed. We therefore account emissions with a stylised marginal emission factor (MEF). The MEF is 394 gCO₂/kWh (combined-cycle gas) in every hour except two cases where it is zero. For South Scotland it is zero when Scotland is nearly carbon-free (below 25 g) while GB-wide intensity exceeds 100 g, a proxy for export-constrained hours; this holds in 27% of hours. For all sites it is zero when GB-wide intensity is below 50 g (4% of hours). Under this accounting the picture changes substantially. The oracle cuts marginal emissions by only 13.5%, and CARMA by 7.3%. The forecast-driven policies are within 7.9–10.7% of the marginal oracle. CARMA is still the best

of them, but its advantage over MPC shrinks to 0.6 points (block-bootstrap CI 0.06–0.91), which is not significant after Holm correction ($p = 0.12$; CARMA better in 17 of 26 weeks). The schedules were optimised for average intensity, not for the MEF. The result nonetheless shows that most of the attributional reductions reported above are reallocations within the GB system. The physical benefit depends on whether shifting absorbs otherwise-curtailed low-carbon output, and a scheduler should be driven by a marginal signal when that is the goal.

5.6 Robustness to prediction errors

We test how CARMA degrades as its predictions deteriorate on every fourth test week of 2024 (13 weeks, $\rho = 0.5$, migration allowed, fresh workload traces). Controlled errors are injected into the two kinds of prediction (Fig. 7). These experiments do not satisfy the assumptions of the bound in Theorem 2, so they probe the degradation profile empirically.

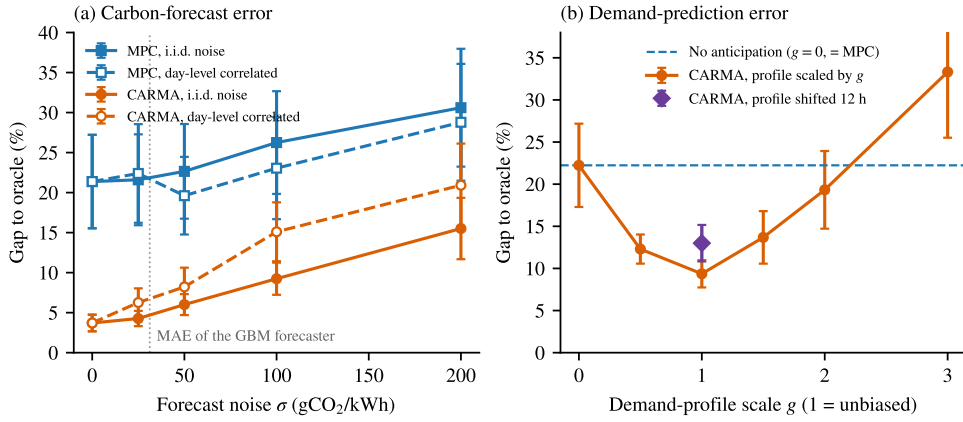


Fig. 7 Degradation under controlled prediction errors (13 test weeks of 2024, $\rho = 0.5$, migration allowed; means with 95% confidence intervals). (a) Carbon-intensity forecasts replaced by the realised values plus Gaussian noise of standard deviation σ . The noise is either independent across issue times and horizons (solid) or shared by all forecasts of the same site and target day (dashed, open markers). (b) Demand profile scaled by g , or shifted by 12 hours so that arrivals are predicted at the wrong time of day.

Carbon-forecast error

With exact forecasts, CARMA’s gap is 3.7% and MPC’s 21.4%. Independent noise is comparatively benign: CARMA’s gap rises smoothly to 4.3%, 6.0%, 9.2% and 15.5% at $\sigma = 25, 50, 100$ and 200 gCO₂/kWh. Noise that persists over a whole target day, as when a weather front is missed, is more damaging. The gaps are then 6.3%, 8.2%, 15.1% and 20.9%, because the planner cannot average out the error by re-planning. MPC deteriorates from 21.4% to 28.8–30.6% at the largest noise level. At every noise level and type, CARMA remains better than MPC with the same forecasts. Only with

the most persistent large errors ($\sigma = 200$) does it fall to the level of MPC with *perfect* forecasts. The actual GBM forecaster (MAE 31 gCO₂/kWh, with errors that are both serially and cross-sectionally correlated) gives a gap of 9.8% in the main experiment. That lies between the independent and correlated cases at comparable error size, so the degradation should not be read as merely i.i.d.

Demand-prediction error

Scaling the learned demand profile by g produces a V-shaped response with its minimum at the unbiased profile ($g = 1$, gap 9.4%). Under-prediction by half ($g = 0.5$) and over-prediction by half ($g = 1.5$) cost 2.9 and 4.3 percentage points. Predicting arrivals at the wrong time of day (a 12-hour shift) costs 3.6 points. Removing anticipation ($g = 0$) returns the MPC level (22.2%). Gross over-prediction ($g = 3$, three times the true demand) is worse than no anticipation (33.3%), because CARMA then holds clean capacity for work that never arrives. A 12-hour shift keeps most of the benefit, which indicates that much of the value comes from anticipating the *volume* of demand rather than its exact timing. Late work was negligible in all 325 runs (at most 0.01% of work). The largest ratio of CARMA’s emissions to the oracle’s over all runs was 1.58.

5.7 Fair carbon attribution

We evaluate Theorem 3 on the 52 test weeks at $\rho = 0.5$, with the four sites as players. For every week and each of the 15 coalitions we compute the hindsight value $c(C)$, and we also run CARMA with only the coalition’s sites and jobs. Pooling pays: the hindsight optimum of the grand coalition emits 3.37 tCO₂ per week, against 6.70 for the four sites operating separately. With CARMA the figures are 3.63 and 6.88. The question is how to share the reduction.

We compare six attributions. The *origin-based* rule charges each site the emissions of its own jobs wherever they run. The *host-based* rule, the analogue of location-based Scope 2 accounting, charges each site the emissions of the work it hosts. The *work-proportional* rule splits the total in proportion to submitted work, and the *stand-alone proportional* rule in proportion to stand-alone emissions. The *Shapley value* [67] averages marginal contributions over all orders of joining. The *dual* rule is Eq. (10). Because $c(C)$ includes the lateness penalty whenever a coalition cannot serve all of its work, we also evaluate an emissions-only game, in which $c(C)$ is the emissions of the hindsight schedule. Stand-alone coalitions leave work unserved in 94% of weeks, although the amount is small. In that game the dual attribution is rescaled to the emissions total and is no longer covered by the theorem. For CARMA’s realised emissions, the dual attribution is applied to the hindsight value and the online excess over the hindsight optimum is split in proportion to work. A check is violated when a coalition is charged more than it would emit on its own.

Table 6 and Fig. 8 report the results. In the hindsight game the dual attribution satisfies all 728 coalition checks, as Theorem 3 guarantees. The dual prices are not always unique, but over all optimal duals (checked in 9 weeks) the attribution of a site varies by at most 0.011 tCO₂ per week, so the choice of dual does not matter in practice. The four accounting rules violate 22–49% of the checks and fail in every week. The origin-based rule is individually irrational for South Scotland in all 52 weeks

Table 6 Coalitional stability of carbon attributions (52 test weeks of 2024, $\rho = 0.5$; one 216-hour arrival episode per week). Checks: share of the 52×14 (episode, proper coalition) pairs in which the coalition is charged more than its stand-alone value. Weeks: share of episodes with at least one violation. IR: share of (episode, site) pairs in which a site is charged more than its stand-alone value. Neg.: share of (episode, site) pairs with a net credit. Max. exc.: largest violation (tCO₂). Panel (a) uses the penalised values $c(C)$ of Theorem 3, panel (b) emissions only, panel (c) CARMA’s realised emissions.

Attribution	Checks (%)	Weeks (%)	IR (%)	Neg. (%)	Max. exc.
<i>(a) Hindsight game, penalised cost $c(C)$ (Theorem 3)</i>					
Origin-based	46.4	100.0	49.5	0.0	1.25
Host-based	49.2	100.0	50.0	0.0	4.99
Work-proportional	44.4	100.0	45.7	0.0	1.43
Stand-alone proportional	22.0	100.0	0.0	0.0	0.84
Shapley	3.7	44.2	0.0	40.9	0.20
Dual	0.0	0.0	0.0	37.5	–
<i>(b) Hindsight game, emissions only (dual rescaled; not covered by Theorem 3)</i>					
Shapley	3.3	42.3	0.0	40.9	0.10
Rescaled dual	0.0	0.0	0.0	37.5	–
<i>(c) Realised emissions of CARMA (online game)</i>					
Origin-based	43.1	100.0	44.2	0.0	1.20
Host-based	47.5	100.0	50.0	0.0	4.78
Work-proportional	43.3	100.0	46.2	0.0	1.48
Stand-alone proportional	21.3	100.0	0.0	0.0	0.86
Shapley	3.8	50.0	0.0	39.9	0.13
Adjusted dual (heuristic)	0.8	11.5	0.0	34.6	0.06

and for North West England in 51: their own jobs are displaced to dirtier capacity by imported work, so they are charged more than they would emit on their own. The host-based rule fails for the same two sites in every week, because they are charged for the imported work they host; North West England, for example, is charged 1.50 tCO₂ per week against 0.59 on its own. Scaling stand-alone emissions down is individually rational by construction, but it still leaves 22% of the coalition checks violated. The Shapley value comes close: it is individually rational and violates 3.7% of the checks (44% of weeks), with a largest excess of 0.20 tCO₂. Its site charges are similar to the dual ones. Both credit the clean sites for the capacity they contribute. Under the dual rule South Scotland receives a net credit of 0.91 tCO₂ per week (Shapley 0.77), North West England is close to neutral, and London’s charge (2.81 t) remains well below its stand-alone emissions (3.91 t). Net credits arise in 38% of the (week, site) pairs. The Shapley value also needs all 2^n coalition values, whereas the dual rule comes from a single solve. The emissions-only game gives almost the same picture: the rescaled dual attribution passes every check, and Shapley fails 3.3%.

With CARMA’s realised emissions, which is the practical case, the adjusted dual attribution remains individually rational for every site in every week. It violates 6 of the 728 coalition checks (0.8%, in 6 of 52 weeks), all involving North West England and

London, joined in five of them by South Scotland, and the largest excess is 0.06 tCO₂. Shapley applied to the online values fails 3.8% of the checks, and the four accounting rules fail 21–48%. Stand-alone coalitions under CARMA leave at most 4.7% of their work late, and unfinished work emits nothing, so the online checks, if anything, favour secession.

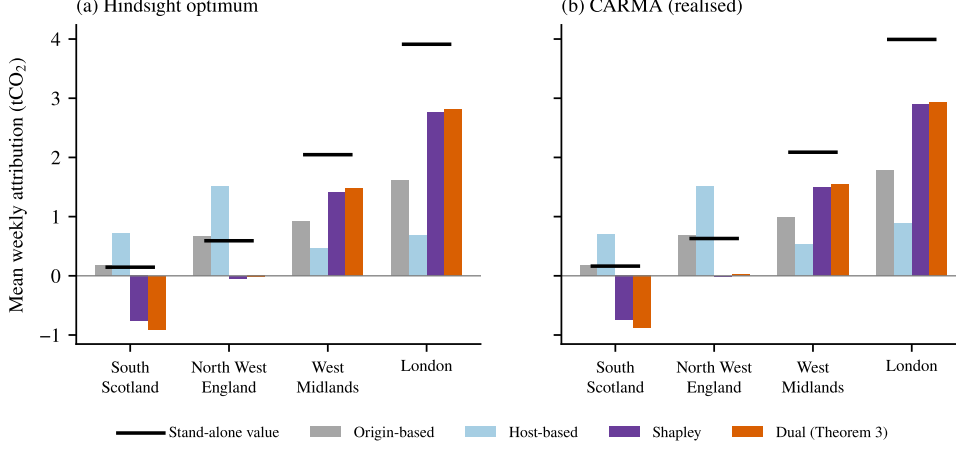


Fig. 8 Mean weekly carbon attribution by site under four rules, compared with each site’s stand-alone value (black lines), 52 test weeks of 2024 at $\rho = 0.5$. (a) Hindsight optimum. (b) CARMA’s realised emissions. Negative values are net credits for contributing low-carbon capacity.

6 Discussion

6.1 Managerial implications

Protect scarce clean capacity

In multi-site carbon-aware scheduling, the largest operational loss does not come from imperfect carbon forecasts. It comes from spending low-carbon capacity on work that could have waited or run elsewhere. Proposition 2 and Theorem 1 show this in the worst case, and the experiments confirm it on average. With migration allowed, a myopic planner with perfect carbon foresight is still 16–21% above the optimum in the synthetic experiment and 33% above it on the Alibaba trace. Clean capacity at a low-carbon site should be treated as a scarce resource with an opportunity cost, as in revenue management. A simple protection level (MPC-Protect) captures part of this value. CARMA captures more, because its ghost jobs price the protection hour by hour from the operator’s own history. The result held when the profile was biased by 39% and demand was bursty.

1427 *Invest in forecasts where they pay*

1428 The value of better intensity forecasts depends on the rest of the policy. For a myopic
1429 multi-site planner, perfect forecasts are worth only 1–3 percentage points; once demand
1430 is anticipated they are worth 4–7 points; without migration they are worth almost the
1431 whole remaining gap. Persistent errors, such as a missed day-ahead wind front, are
1432 more costly than independent ones. Forecasting effort, for example the integration of
1433 numerical weather predictions [43, 44], should therefore target day-level biases, and in
1434 multi-site settings it should follow the fix of demand myopia.

1435

1436 *Use risk premia with care*

1437 The conformal cost views help a myopic planner but not CARMA. They work by
1438 making volatile sites and distant hours look dirtier, which crudely imitates anticipation.
1439 Once demand is anticipated, the same distortion gives up useful deferrals. Calibrated
1440 quantiles are better suited to reporting and to service-level decisions than to reshaping
1441 costs by default.

1442

1443 *Emissions or service near saturation*

1444 Without migration and at high load, every carbon-aware policy delayed work, and
1445 under a service-adjusted metric carbon-agnostic operation was preferable. Operators
1446 should combine temporal shifting with explicit service constraints, or restrict it, when
1447 local capacity is close to saturation.

1448

1449 *Attributional versus physical savings*

1450 The headline reductions are attributional: they measure how much of the work runs
1451 where and when the *average* regional mix is clean. Within a single market such as
1452 Great Britain, moving load between regions mostly reassigns the average mix. Under
1453 a stylised marginal accounting, the achievable reduction fell from 58% to 14% and
1454 CARMA’s advantage over MPC became small. Carbon-aware shifting is physically
1455 most valuable when it absorbs low-carbon output that would otherwise be curtailed,
1456 for example behind transmission constraints between Scotland and England, when
1457 temporal shifts change which plant is marginal, or when sites lie in different markets.
1458 Our marginal accounting is a stylised proxy, not an estimate of GB-wide abatement,
1459 so it indicates the direction and rough size of this effect rather than its value. In such
1460 settings a marginal signal should drive the scheduler. The scheduling framework is
1461 unchanged: only the cost coefficients of the flow network change. Idle power works in
1462 the same direction. If servers stay powered, deferral saves only their dynamic energy,
1463 and carbon-aware scheduling should be paired with right-sizing [18].

1464

1465 *Fair internal carbon accounting*

1466 Pooling halved the hindsight emissions of the four sites. However, origin-based account-
1467 ing, which charges each site for its own jobs wherever they run, would have made South
1468 Scotland worse off than on its own in every week. A pool of organisations or business
1469 units allocated this way is unstable, because its cleanest members have an incentive to
1470 leave. The dual attribution of Theorem 3 charges each site the carbon shadow price

1471

of its demand and credits it for the capacity it contributes. It is budget-balanced, no coalition would prefer to secede, and it is a by-product of the scheduling problem. It suits internal carbon pricing and the sharing of the benefits of carbon-aware scheduling among business units. It is not a substitute for external GHG Protocol reporting, where negative Scope 2 emissions do not exist [66]. The players must also be entities that could credibly schedule on their own, such as separate operators, business units with their own capacity, or tenants with reserved shares. Because credits reward contributed capacity, capacity declarations should be audited.

Deployment

CARMA needs a carbon-intensity feed (average or marginal), a demand profile computed from the operator’s own job logs and a network-flow solver. It decides within tens of milliseconds and returns a plan that is optimal for its model, which makes its decisions auditable. Absolute savings scale with the fleet: for the 1,700-server fleet simulated here, CARMA saves about 16 tCO₂ per year relative to myopic control and about 165 tCO₂ relative to carbon-agnostic operation, in attributional terms.

6.2 Limitations and future research

The study has several limitations. First, the synthetic workloads use a profile learned from the same generator, which favours anticipation. The trace experiment partly addresses this concern, but it covers only four evaluation days of one cluster, arrival times are first task starts, work is based on planned CPU requests, deadlines and origins are synthetic, and jobs are batched. Its 156 episodes vary carbon conditions but share one demand chronology. Second, the regional intensities published by NESO are modelled estimates, used here as realised values, 52 hours of them are missing and filled by persistence, and the marginal analysis relies on a stylised proxy with fixed thresholds, not on a dispatch model or observed curtailment. Third, the facility parameters (PUE, network energy, a 96-core server) are stylised, and embodied emissions are outside the scope except as the floor that makes θ finite. Fourth, the forecaster excludes weather inputs, so its day-ahead skill is limited. Fifth, bag-of-tasks jobs with free preemption and migration are an idealisation: jobs with state, data gravity, bandwidth limits or data-sovereignty constraints would add restrictions that break the pure network structure. Sixth, the guarantees of Theorem 2 assume a backstop resource, a positive emission floor and an episode within the planning window, none of which holds in the simulations. The theorem describes the mechanism and the experiments its empirical profile. Its robustness term is shared by all on-time policies and is far from the $\Omega(\sqrt{\theta})$ lower bound. Seventh, the core property is exact only for the hindsight benchmark.

Several extensions follow naturally. The ghost-demand profile could be replaced by probabilistic arrival forecasts, with the anticipation weight set by a newsvendor-type argument, or by scenario-based planning that treats demand and carbon uncertainty jointly [57]. Tightening robustness towards the lower bound while retaining consistency, for example by protecting deferral decisions with reservation prices [24, 26], is an open theoretical question. The framework can also be driven by marginal emission signals, coupled with right-sizing, on-site renewables and storage [21, 22] and demand-response programmes [23], and evaluated across several electricity markets.

1519 7 Conclusions

1520
1521 This paper studied carbon-aware scheduling of deferrable computing workloads across
1522 data-centre sites as a sustainable-operations problem. We showed that the offline
1523 problem, including width limits, deadlines and the emissions of data transfer, is a
1524 minimum-cost flow, which yields exact oracles and fast online planning. We proved that
1525 uncertainty about demand alone forces every online algorithm to a competitive ratio of
1526 order $\sqrt{\theta}$, even with perfect carbon forecasts, and that myopic planning does far worse.
1527 We proposed CARMA, which adds ghost jobs for expected arrivals to the planning
1528 network. Given backstop capacity, it meets every deadline under any predictions, is
1529 optimal under exact predictions and degrades linearly with prediction error. The dual
1530 prices of the pooled problem yield a carbon attribution that is budget-balanced and in
1531 the core of the pooling game.

1532 In 52 out-of-sample weeks of 2024 on British regional grid data, with migration
1533 allowed, CARMA cut attributional emissions relative to carbon-agnostic operation by
1534 36–68%. It came within 8.0–9.8% of the clairvoyant optimum, against 15–16% for the
1535 best of the myopic, protection-level and risk-averse baselines, and it did so in almost
1536 every week. On a real cluster trace, with a demand profile biased by 39% and bursty
1537 arrivals, its gap was 20.7% against 26.7% for the best baseline. Anticipating demand was
1538 worth far more than perfect carbon foresight, and correlated day-level forecast errors
1539 hurt more than independent ones. Three results qualify these gains. Without migration,
1540 reductions are small (9–18%), and near saturation carbon-aware shifting delays work
1541 enough that carbon-agnostic operation is preferable under a service-adjusted metric.
1542 Under a stylised marginal-emission accounting, the achievable reduction fell to 14%
1543 and CARMA’s advantage over MPC was no longer significant, so the physical value of
1544 shifting within one grid depends on absorbing otherwise-curtailed low-carbon output.
1545 With always-on servers, most emissions no longer depend on the schedule.

1546 Origin-based and host-based carbon accounting violated coalitional stability in
1547 every test week. The dual attribution never did in hindsight and failed 6 of 728 checks,
1548 by at most 0.06 tCO₂, for CARMA’s realised emissions. Its charges were close to the
1549 Shapley value, which it obtains from a single solve. Carbon-aware computing is most
1550 effective when schedulers plan for the work that has not yet arrived, not only for the
1551 grid conditions that lie ahead, and when its benefits are measured and shared on a
1552 sound basis.

1553

1554 Declarations

1555

1556 **Funding.** This research did not receive any specific grant from funding agencies in
1557 the public, commercial, or not-for-profit sectors.

1558 **Competing interests.** The author declares no competing interests.

1559

1560 **Ethics approval and consent to participate.** Not applicable.

1561 **Consent for publication.** Not applicable.

1562

1563 **Data availability.** The carbon-intensity data are publicly available from the NESO
1564 Carbon Intensity API [45] under the CC BY 4.0 licence. The Alibaba 2018 cluster

trace is publicly available [63]. The data snapshot used in this study, the processed trace and all raw experiment outputs are available from the author's repository (<https://github.com/vinhqdang/Sustainable-Operations-and-Computers>).

Materials availability. Not applicable.

Code availability. All code, including the workload generator with fixed random seeds, is available from the same repository.

Author contribution. Q.-V.D. (ORCID 0000-0002-3877-8024): conceptualization, methodology, software, data curation, formal analysis, investigation, validation, visualization, writing – original draft, writing – review and editing.

Appendix A Proofs

Atoms

Splitting a job j into sub-jobs $j_1, \dots, j_k$ with work w_{j_i} and widths $m_{j_i} = m_j w_{j_i} / w_j$ does not change the optimal value of any planning problem. Given a solution for the merged job, the allocation $x_{j_i s}(t) = x_{js}(t) w_{j_i} / w_j$ satisfies every constraint of the split problem at the same cost. Conversely, summing the sub-job allocations gives a feasible allocation for the merged job, because $\sum_i m_{j_i} = m_j$. Splitting also preserves window-feasibility, since $w_{j_i} / m_{j_i} = w_j / m_j$. We may therefore compare true and predicted job sets atom by atom.

A.1 Proof of Theorem 1

Normalise $L = 1$, so $U = \theta$. Site A has one server with cost $p \in [1, \theta]$ in hour 0 and cost 1 in hour 1. Site B has ample capacity at cost θ in both hours. Transfer costs are zero. A flexible job ($w = m = 1$, window $\{0, 1\}$) is released at $t = 0$. The adversary either releases nothing more (scenario 1) or releases an urgent job ($w = m = 1$, window $\{1\}$) at $t = 1$ (scenario 2). All costs are known in advance.

Consider any deterministic online algorithm that may split work fractionally, and let $x \in [0, 1]$ be the amount of flexible work it runs at $t = 0$. Running at B in hour 0 costs $\theta \geq p$ and is dominated, so we may assume this work runs at A . In scenario 1 the algorithm finishes the remaining $1 - x$ at A in hour 1, so $\text{ALG}_1 = xp + (1 - x)$, while $\text{OPT}_1 = 1$. In scenario 2, hour 1 must process $2 - x$ units and A can take only one of them, so $\text{ALG}_2 = xp + 1 + (1 - x)\theta$, while $\text{OPT}_2 = p + 1$. The ratio $r_1(x) = 1 + x(p - 1)$ increases in x , and the ratio $r_2(x) = (1 + \theta - x(\theta - p)) / (p + 1)$ decreases in x . Hence $\min_x \max\{r_1, r_2\}$ is attained where they are equal, at $x^* = (\theta - p) / (p^2 - p - 1 + \theta) \in [0, 1]$, and its value is

$$1 + \frac{(p - 1)(\theta - p)}{p^2 - p - 1 + \theta}.$$

The adversary chooses p to maximise this value, which gives $R^*(\theta)$. The choice $p = \sqrt{\theta}$ gives the stated closed-form lower bound. For a randomised algorithm, the expected cost in each scenario is affine in x , so it equals the cost of the deterministic fractional algorithm that plays $\mathbb{E}[x]$, and the bound applies against an oblivious adversary. Myopic MPC sees only the flexible job at $t = 0$ and, for any $p > 1$, defers it entirely to the

cheaper hour 1 ($x = 0$). Its ratio in scenario 2 is $(1 + \theta)/(p + 1)$, which tends to $(1 + \theta)/2$ as $p \rightarrow 1^+$, so its competitive ratio is at least $(1 + \theta)/2$. $\square$

1613

1614 A.2 Proof of Proposition 3 and Theorem 2

1615

1616 *Proposition 3*

1617 A policy that completes all work W on time executes every server-hour at a unit cost of
1618 at most U , so its emissions are at most UW . Every server-hour in the oracle's solution
1619 costs at least L , and slack costs $\Pi > U \geq L$, so $\text{OPT} \geq LW$. $\square$

1620

1621 *Deadline safety*

1622 We first show by induction that every pending real job j is window-feasible at the start
1623 of every hour: its remaining work satisfies $r_j \leq m_j(d_j - t)$. This holds at release by
1624 (A1). Suppose it holds at hour t , and suppose the plan of CARMA leaves some real
1625 work $u_j > 0$ unfinished. Because $r_j \leq m_j(d_j - t)$, some hour $\tau \in [t, d_j)$ carries less
1626 than m_j units of job j in the plan. Moving a small amount of u_j to the backstop in
1627 hour τ keeps the plan feasible, because the backstop has unlimited capacity. It changes
1628 the planned cost by a negative amount, because the planner prices the backstop at
1629 most at $U < \Pi$ (A1), whatever its forecasts of the other costs are. Since $d_j - t \leq H$
1630 (A1), hour τ lies inside the planning window. This contradicts optimality, so the plan
1631 completes all real work. After the first hour is executed, the rest of the plan is a feasible
1632 completion, so $r'_j \leq m_j(d_j - t - 1)$ and the induction step holds. No real job is ever
1633 late, whatever the ghost jobs are, and θ -competitiveness follows from Proposition 3.

1634

1635 *Telescoping*

1636 Let σ_t be the state at the start of hour t : the remaining work of all real jobs released
1637 by t . Let $V_t(\sigma, G; c)$ be the optimal value of (2)–(5) over hours $t, \dots, T - 1$ for the
1638 pending jobs σ and the future jobs G under costs c . Write $\bar{V}_t = V_t(\sigma_t, F_t; c)$, so that
1639 $\bar{V}_0 = \text{OPT}$ and $\bar{V}_T = 0$. For a feasible hour- t allocation x with cost $g_t(x)$ and successor
1640 state $\sigma^+(x)$, define

1641

$$1642 \quad Q_t(x) = g_t(x) + V_{t+1}(\sigma^+(x), F_t; c), \quad \hat{Q}_t(x) = g_t(x) + V_{t+1}(\sigma^+(x), \hat{F}_t; \hat{c}_t).$$

1643

1644 By dynamic programming over a deterministic future, $\min_x Q_t(x) = \bar{V}_t$, and $Q_t(x_t) =$
1645 $g_t(x_t) + \bar{V}_{t+1}$ for the executed action x_t . Summing over t gives

1646

$$1647 \quad \text{ALG} - \text{OPT} = \sum_{t=0}^{T-1} \delta_t, \quad \delta_t = Q_t(x_t) - \min_x Q_t(x) \geq 0.$$

1648

1649 Ghost jobs are released at $t + 1$ or later, the planning window reaches T by (A2),
1650 and hour- t costs are observed. The planning problem solved by CARMA at hour t is
1651 therefore exactly $\min_x \hat{Q}_t(x)$, and x_t is a minimiser.

1652

1653

1654

1655

1656

One-step loss

Let $x^* \in \arg \min Q_t$. Since $\hat{Q}_t(x_t) \leq \hat{Q}_t(x^*)$, we have $\delta_t \leq \Delta(x_t) - \Delta(x^*)$, where $\Delta = Q_t - \hat{Q}_t$. Write $\Delta = D_1 + D_2$ with

$$\begin{aligned} D_1(x) &= V_{t+1}(\sigma^+(x), F_t; c) - V_{t+1}(\sigma^+(x), \hat{F}_t; c), \\ D_2(x) &= V_{t+1}(\sigma^+(x), \hat{F}_t; c) - V_{t+1}(\sigma^+(x), \hat{F}_t; \hat{c}_t). \end{aligned}$$

Cost error. An optimal solution under $\hat{c}_t$ is feasible under c , and its cost changes by at most ε_t per unit of flow on future-hour arcs. The same holds with the roles of c and $\hat{c}_t$ exchanged. Hence $|D_2(x)| \leq \varepsilon_t W_t$ for all x , and $D_2(x_t) - D_2(x^*) \leq 2\varepsilon_t W_t$.

Demand error. Under (A1), the penalty attached to a window-feasible job does not affect any optimal value, provided it is at least U . If a solution leaves slack on such a job, the argument used for robustness moves that slack to the backstop at no greater cost. True future jobs (penalty Π) and ghost jobs (penalty Π_g) can therefore be treated as identical atoms whenever they coincide. Let a be a window-feasible atom and G any set of future jobs. For every state σ ,

$$Lw_a \leq V(\sigma, G \cup \{a\}) - V(\sigma, G) \leq Uw_a.$$

For the upper bound, add a to an optimal solution for G by running it on the backstop at rate $w_a/(d_a - a_a) \leq m_a$ in every hour of its window. For the lower bound, delete a 's flow and slack from an optimal solution for $G \cup \{a\}$; each deleted unit cost at least $\min\{L, \Pi_a\} = L$, since $\Pi_a \geq U \geq L$. Now transform $\hat{F}_t$ into F_t by removing $\hat{F}_t \setminus F_t$ and adding $F_t \setminus \hat{F}_t$, one atom at a time. Then $D_1(x)$ is a sum of increments, each lying in $[Lw_a, Uw_a]$ (with a minus sign for removals) for every x . Therefore $D_1(x_t) - D_1(x^*) \leq (U - L) \sum_a w_a = (U - L) d_t$.

Combining the two bounds gives $\delta_t \leq (U - L)d_t + 2\varepsilon_t W_t$. Summing over t and taking the minimum with the robustness bound proves (9). If $d_t = \varepsilon_t = 0$ for all t , then $\delta_t = 0$ and $\text{ALG} = \text{OPT}$. $\square$

A.3 Proof of Theorem 3

Write the problem of coalition C as $c(C) = \min\{c^\top z : A_\leq z = b_\leq(C), A_\leq z \leq b_\leq(C), z \geq 0\}$, where $z = (x, u)$. Jobs submitted outside C have zero work and zero width, and sites outside C have zero capacity. This forces the corresponding flows to zero, so the problem coincides with the stand-alone problem of C . The right-hand side is additive over sites: $b(C) = \sum_{o \in C} b^o$, where b^o collects the work and widths of the jobs submitted at o and the capacity of site o . The matrix A and the cost c do not depend on C . The dual problem $\max\{b(C)^\top y : A^\top y \leq c, y_\leq \leq 0\}$ therefore has a feasible set that is the same for all coalitions.

Let $y^* = (\lambda, \nu, \mu)$ be an optimal dual solution for $\mathcal{S}$. The slack variables make every primal problem feasible, and costs are non-negative, so strong duality gives $c(\mathcal{S}) = b(\mathcal{S})^\top y^* = \sum_o (b^o)^\top y^* = \sum_o \phi_o$, which is budget balance. For any coalition C , y^* is dual-feasible for C , so weak duality gives $c(C) \geq b(C)^\top y^* = \sum_{o \in C} \phi_o$, which is the core condition [31]. Expanding $(b^o)^\top y^*$ gives (10). By Proposition 1 the problem

is a min-cost flow, so optimal node potentials, together with the reduced costs they induce on saturated capacity arcs, also give an optimal dual solution. Any optimal dual solution can be used; in our experiments it is returned by the LP solver HiGHS. $\square$

Appendix B Forecast evaluation

Table B.1 and Fig. B.1a summarise forecast accuracy in 2024; hours affected by gap filling are excluded. Averaged over horizons and sites, the direct GBM forecaster has a mean absolute error (MAE) of 31.7 gCO₂/kWh, against 35.3 for persistence and 40.9 for the seasonal-naive forecast. The advantage is largest at short and medium horizons: MAE falls by 23% at 1 hour (6.4 against 8.3) and by 24% at 6 hours (23.6 against 31.2). The advantage shrinks with the horizon, and at 24 hours the GBM is worse than both benchmarks (42.7 against 40.9). Without weather forecasts, day-ahead changes in wind output are hard to predict from the intensity history alone [44]. By site, the improvement over persistence is largest in absolute terms for the volatile southern sites (West Midlands 50.2 against 57.7, London 38.0 against 43.4), although their MAE is also the highest. In South Scotland, where intensity is low and often flat, persistence is better overall (14.5 against 16.2). Diebold–Mariano tests [68] with absolute-error loss and HAC variance confirm these patterns. For the three English sites, the GBM is significantly more accurate than persistence at horizons of 1–18 hours ($p \leq 0.002$) and indistinguishable at 24 hours ($p \geq 0.09$). For South Scotland it is significantly better at 1 and 3 hours, indistinguishable at 6 hours and significantly worse from 12 hours onwards.

Table B.1 Forecast accuracy on the 2024 test year (MAE in gCO₂/kWh; mean over sites unless stated).

Model	Horizon h (hours)					All horizons	
	1	3	6	12	24	MAE	RMSE
Persistence	8.3	20.0	31.2	38.6	40.9	35.3	49.2
Seasonal naive (24 h)	40.9	40.9	40.8	40.8	40.9	40.9	57.4
Direct GBM (this study)	6.4	15.1	23.6	34.5	42.7	31.7	41.6

MAE by site over all horizons (persistence / GBM): South Scotland 14.5 / 16.2; North West England 25.5 / 22.3; West Midlands 57.7 / 50.2; London 43.4 / 38.0.

Calibration matters more than point accuracy for the risk-adjusted cost view, and Fig. B.1b shows the effect of distribution shift. Static split-conformal bounds, calibrated on July–September 2023, under-cover in 2024 at every level: at $\beta = 0.9$, annual coverage was 0.868 and monthly coverage ranged from 0.78 to 0.95. With the adaptive recalibration, annual coverage is within 0.003 of the target at all levels (0.499, 0.598, 0.697, 0.797 and 0.898). Monthly coverage stays within about four percentage points of the target (at most 4.2 points), and coverage is uniform across horizons. At $\beta = 0.5$, the recalibrated median lies on average 10.3 gCO₂/kWh below the point forecast, which reflects the downward drift in intensity that the static model does not

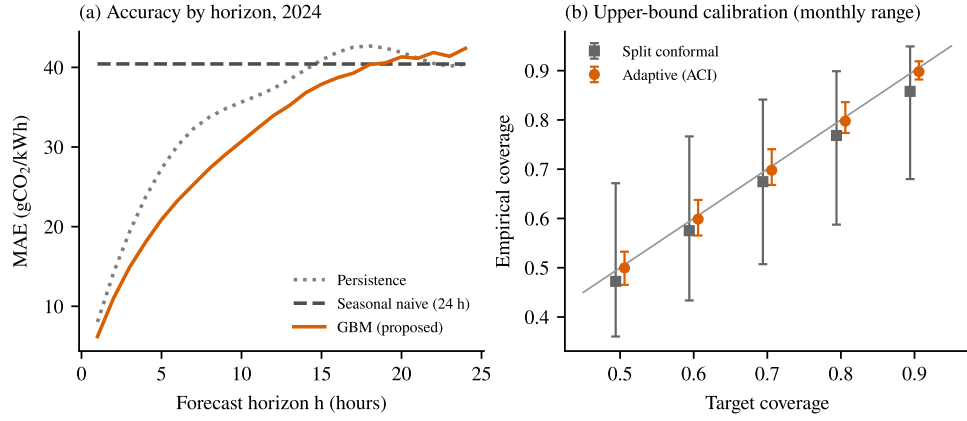


Fig. B.1 Forecast evaluation in 2024. (a) MAE by horizon, averaged over sites. (b) Empirical coverage of one-sided upper bounds U^β against the target level β for static split conformal calibration and adaptive conformal inference (ACI). Markers show the annual coverage and bars the range across the 12 months.

capture. Coverage is not the whole story. In pinball loss, the adaptive bounds improve on static calibration at levels 0.5–0.7 (15.01 against 15.16 at 0.5) but are worse at 0.8 and 0.9 (9.07 against 8.36 at 0.9). Delayed feedback explains much of this. The level for horizon h is updated only when the outcome is observed, h hours after issue, so a day of under-forecasts drives the level of long horizons towards 1 before any correction arrives. At $\beta = 0.9$, 5.6% of the 2024 bounds (0.5% at $\beta = 0.8$) were at the fallback ceiling, the training-period maximum of the site. These features, together with the finite ceiling and the quantile grid, are why we treat the procedure as an empirical recalibration and do not invoke the coverage guarantee of ACI [54]. Annual coverage says nothing about conditional coverage in the site-hours that the optimiser selects.

Appendix C Numerical check of Theorem 2

(Results pending.)

References

- [1] Masanet, E., Shehabi, A., Lei, N., Smith, S., Koomey, J.: Recalibrating global data center energy-use estimates. *Science* **367**(6481), 984–986 (2020) <https://doi.org/10.1126/science.aba3758>
- [2] Shehabi, A., Smith, S., Sartor, D., Brown, R., Herrlin, M., Koomey, J., Masanet, E., Horner, N., Azevedo, I., Lintner, W.: United States Data Center Energy Usage Report. Technical report, Lawrence Berkeley National Laboratory, Berkeley, CA (2016). <https://doi.org/10.2172/1372902>
- [3] Freitag, C., Berners-Lee, M., Widdicks, K., Knowles, B., Blair, G.S., Friday, A.:

1795 The real climate and transformative impact of ICT: A critique of estimates, trends,
1796 and regulations. *Patterns* **2**(9), 100340 (2021) [https://doi.org/10.1016/j.patter.](https://doi.org/10.1016/j.patter.2021.100340)
1797 [2021.100340](https://doi.org/10.1016/j.patter.2021.100340)
1798
1799 [4] Strubell, E., Ganesh, A., McCallum, A.: Energy and Policy Considerations for Deep
1800 Learning in NLP. In: Proceedings of the 57th Annual Meeting of the Association
1801 for Computational Linguistics, pp. 3645–3650. Association for Computational
1802 Linguistics, Stroudsburg, PA, USA (2019). <https://doi.org/10.18653/v1/p19-1355>
1803
1804 [5] Patterson, D., Gonzalez, J., Hölzle, U., Le, Q., Liang, C., Munguia, L.-M.,
1805 Rothchild, D., So, D.R., Texier, M., Dean, J.: The Carbon Footprint of Machine
1806 Learning Training Will Plateau, Then Shrink. *Computer* **55**(7), 18–28 (2022)
1807 <https://doi.org/10.1109/mc.2022.3148714>
1808
1809 [6] Dodge, J., Prewitt, T., Combes, R., Odmark, E., Schwartz, R., Strubell, E., Luc-
1810 cioni, A.S., Smith, N.A., DeCario, N., Buchanan, W.: Measuring the Carbon
1811 Intensity of AI in Cloud Instances. In: 2022 ACM Conference on Fairness Account-
1812 ability and Transparency, pp. 1877–1894. ACM, New York, NY, USA (2022).
1813 <https://doi.org/10.1145/3531146.3533234>
1814
1815 [7] Staffell, I.: Measuring the progress and impacts of decarbonising British electricity.
1816 *Energy Policy* **102**, 463–475 (2017) <https://doi.org/10.1016/j.enpol.2016.12.037>
1817
1818 [8] Radovanović, A., Koningstein, R., Schneider, I., Chen, B., Duarte, A., Roy, B.,
1819 Xiao, D., Haridasan, M., Hung, P., Care, N., Talukdar, S., Mullen, E., Smith,
1820 K., Cottman, M., Cirne, W.: Carbon-Aware Computing for Datacenters. *IEEE*
1821 *Transactions on Power Systems* **38**(2), 1270–1280 (2023) [https://doi.org/10.1109/](https://doi.org/10.1109/tpwrs.2022.3173250)
1822 [tpwrs.2022.3173250](https://doi.org/10.1109/tpwrs.2022.3173250)
1823
1824 [9] Bashir, N., Guo, T., Hajiesmaili, M., Irwin, D., Shenoy, P., Sitaraman, R., Souza, A.,
1825 Wierman, A.: Enabling Sustainable Clouds: The Case for Virtualizing the Energy
1826 System. In: Proceedings of the ACM Symposium on Cloud Computing, pp. 350–358.
1827 ACM, New York, NY, USA (2021). <https://doi.org/10.1145/3472883.3487009>
1828
1829 [10] Wiesner, P., Behnke, I., Scheinert, D., Gontarska, K., Thamsen, L.: Let’s wait
1830 awhile: How temporal workload shifting can reduce carbon emissions in the cloud.
1831 In: Proceedings of the 22nd International Middleware Conference, pp. 260–272.
1832 ACM, New York, NY, USA (2021). <https://doi.org/10.1145/3464298.3493399>
1833
1834 [11] Goiri, Í., Le, K., Nguyen, T.D., Guitart, J., Torres, J., Bianchini, R.: GreenHadoop:
1835 Leveraging green energy in data-processing frameworks. In: Proceedings of the
1836 7th ACM European Conference on Computer Systems, pp. 57–70. ACM, New
1837 York, NY, USA (2012). <https://doi.org/10.1145/2168836.2168843>
1838
1839 [12] Hanafy, W.A., Liang, Q., Bashir, N., Irwin, D., Shenoy, P.: CarbonScaler: Leverag-
1840 ing Cloud Workload Elasticity for Optimizing Carbon-Efficiency. *Proceedings of*
the ACM on Measurement and Analysis of Computing Systems **7**(3), 1–28 (2023)

- <https://doi.org/10.1145/3626788> 1841
- [13] Liu, Z., Lin, M., Wierman, A., Low, S.H., Andrew, L.L.H.: Greening geographical load balancing. In: Proceedings of the ACM SIGMETRICS Joint International Conference on Measurement and Modeling of Computer Systems, pp. 233–244. ACM, New York, NY, USA (2011). <https://doi.org/10.1145/1993744.1993767> 1842–1846
- [14] Zheng, J., Chien, A.A., Suh, S.: Mitigating Curtailment and Carbon Emissions through Load Migration between Data Centers. *Joule* **4**(10), 2208–2222 (2020) <https://doi.org/10.1016/j.joule.2020.08.001> 1847–1850
- [15] Sukprasert, T., Souza, A., Bashir, N., Irwin, D., Shenoy, P.: On the Limitations of Carbon-Aware Temporal and Spatial Workload Shifting in the Cloud. In: Proceedings of the Nineteenth European Conference on Computer Systems, pp. 924–941. ACM, New York, NY, USA (2024). <https://doi.org/10.1145/3627703.3650079> 1851–1856
- [16] Goiri, Í., Le, K., Haque, M.E., Beauchea, R., Nguyen, T.D., Guitart, J., Torres, J., Bianchini, R.: GreenSlot: Scheduling energy consumption in green datacenters. In: Proceedings of 2011 International Conference for High Performance Computing, Networking, Storage and Analysis, pp. 1–11. ACM, New York, NY, USA (2011). <https://doi.org/10.1145/2063384.2063411> 1857–1862
- [17] Rao, L., Liu, X., Xie, L., Liu, W.: Minimizing Electricity Cost: Optimization of Distributed Internet Data Centers in a Multi-Electricity-Market Environment. In: 2010 Proceedings IEEE INFOCOM, pp. 1–9. IEEE, Piscataway, NJ, USA (2010). <https://doi.org/10.1109/infcom.2010.5461933> 1863–1866
- [18] Lin, M., Wierman, A., Andrew, L.L.H., Thereska, E.: Dynamic Right-Sizing for Power-Proportional Data Centers. *IEEE/ACM Transactions on Networking* **21**(5), 1378–1391 (2013) <https://doi.org/10.1109/tnet.2012.2226216> 1867–1871
- [19] Wiesner, P., Scheinert, D., Wittkopp, T., Thamsen, L., Kao, O.: Cucumber: Renewable-aware admission control for delay-tolerant cloud and edge workloads. In: Euro-Par 2022: Parallel Processing. Lecture Notes in Computer Science, pp. 218–232. Springer, New York, NY, USA (2022). https://doi.org/10.1007/978-3-031-12597-3_14 1872–1876
- [20] Gsteiger, V.U., Long, P.H.D., Sun, Y.J., Javanrood, P., Shahrad, M.: Caribou: Fine-grained geospatial shifting of serverless applications for sustainability. In: Proceedings of the ACM SIGOPS 30th Symposium on Operating Systems Principles, pp. 403–420. ACM, New York, NY, USA (2024). <https://doi.org/10.1145/3694715.3695954> 1877–1882
- [21] Souza, A., Bashir, N., Murillo, J., Hanafy, W., Liang, Q., Irwin, D., Shenoy, P.: Ecovisor: A Virtual Energy System for Carbon-Efficient Applications. In: Proceedings of the 28th ACM International Conference on Architectural Support 1883–1886

- for Programming Languages and Operating Systems, Volume 2, pp. 252–265. ACM, New York, NY, USA (2023). <https://doi.org/10.1145/3575693.3575709>
- [22] Acun, B., Lee, B., Kazhamiaka, F., Maeng, K., Gupta, U., Chakkaravarthy, M., Brooks, D., Wu, C.-J.: Carbon Explorer: A Holistic Framework for Designing Carbon Aware Datacenters. In: Proceedings of the 28th ACM International Conference on Architectural Support for Programming Languages and Operating Systems, Volume 2, pp. 118–132. ACM, New York, NY, USA (2023). <https://doi.org/10.1145/3575693.3575754>
- [23] Wierman, A., Liu, Z., Liu, I., Mohsenian-Rad, H.: Opportunities and challenges for data center demand response. In: International Green Computing Conference, pp. 1–10. IEEE, Piscataway, NJ, USA (2014). <https://doi.org/10.1109/igcc.2014.7039172>
- [24] El-Yaniv, R., Fiat, A., Karp, R.M., Turpin, G.: Optimal search and one-way trading online algorithms. *Algorithmica* **30**(1), 101–139 (2001) <https://doi.org/10.1007/s00453-001-0003-0>
- [25] Lechowicz, A., Christianson, N., Zuo, J., Bashir, N., Hajiesmaili, M., Wierman, A., Shenoy, P.: The online pause and resume problem: Optimal algorithms and an application to carbon-aware load shifting. *Proceedings of the ACM on Measurement and Analysis of Computing Systems* **7**(3), 1–32 (2023) <https://doi.org/10.1145/3626776>
- [26] Lechowicz, A., Christianson, N., Sun, B., Bashir, N., Hajiesmaili, M., Wierman, A., Shenoy, P.: Learning-augmented competitive algorithms for spatiotemporal online allocation with deadline constraints. *Proceedings of the ACM on Measurement and Analysis of Computing Systems* **9**(1), 1–49 (2025) <https://doi.org/10.1145/3711701>
- [27] Lykouris, T., Vassilvitskii, S.: Competitive caching with machine learned advice. *Journal of the ACM* **68**(4), 1–25 (2021) <https://doi.org/10.1145/3447579>
- [28] Jasin, S., Kumar, S.: A re-solving heuristic with bounded revenue loss for network revenue management with customer choice. *Mathematics of Operations Research* **37**(2), 313–345 (2012) <https://doi.org/10.1287/moor.1120.0537>
- [29] Horn, W.A.: Some simple scheduling algorithms. *Naval Research Logistics Quarterly* **21**(1), 177–185 (1974) <https://doi.org/10.1002/nav.3800210113>
- [30] Federgruen, A., Groenevelt, H.: Preemptive scheduling of uniform machines by ordinary network flow techniques. *Management Science* **32**(3), 341–349 (1986) <https://doi.org/10.1287/mnsc.32.3.341>
- [31] Owen, G.: On the core of linear production games. *Mathematical Programming* **9**(1), 358–370 (1975) <https://doi.org/10.1007/bf01681356>

- [32] Kalai, E., Zemel, E.: Totally balanced games and games of flow. *Mathematics of Operations Research* **7**(3), 476–478 (1982) <https://doi.org/10.1287/moor.7.3.476>
- [33] Gahm, C., Denz, F., Dirr, M., Tuma, A.: Energy-efficient scheduling in manufacturing companies: A review and research framework. *European Journal of Operational Research* **248**(3), 744–757 (2016) <https://doi.org/10.1016/j.ejor.2015.07.017>
- [34] Asadpour, M., Hodaei, Z., Azami, M., Kehtari, E., Vesal, N.: A green model for identical parallel machines scheduling problem considering tardy jobs and job splitting property. *Sustainable Operations and Computers* **3**, 149–155 (2022) <https://doi.org/10.1016/j.susoc.2022.01.002>
- [35] Sharma, S., Rawat, P.S.: Efficient resource allocation in cloud environment using SHO-ANN-based hybrid approach. *Sustainable Operations and Computers* **5**, 141–155 (2024) <https://doi.org/10.1016/j.susoc.2024.07.001>
- [36] Ahuja, R.K., Magnanti, T.L., Orlin, J.B.: *Network Flows: Theory, Algorithms, and Applications*. Prentice Hall, Englewood Cliffs, NJ (1993)
- [37] Orlin, J.B.: A polynomial time primal network simplex algorithm for minimum cost flows. *Mathematical Programming* **78**(2), 109–129 (1997) <https://doi.org/10.1007/bf02614365>
- [38] Powell, W.B.: A unified framework for stochastic optimization. *European Journal of Operational Research* **275**(3), 795–821 (2019) <https://doi.org/10.1016/j.ejor.2018.07.014>
- [39] Rawlings, J.B., Mayne, D.Q., Diehl, M.M.: *Model Predictive Control: Theory, Computation, and Design*, 2nd edn. Nob Hill Publishing, Madison, WI, USA (2017)
- [40] Mayne, D.Q., Rawlings, J.B., Rao, C.V., Scokaert, P.O.M.: Constrained model predictive control: Stability and optimality. *Automatica* **36**(6), 789–814 (2000) [https://doi.org/10.1016/S0005-1098\(99\)00214-9](https://doi.org/10.1016/S0005-1098(99)00214-9)
- [41] Bokde, N.D., Tranberg, B., Andresen, G.B.: Short-term CO₂ emissions forecasting based on decomposition approaches and its impact on electricity market scheduling. *Applied Energy* **281**, 116061 (2021) <https://doi.org/10.1016/j.apenergy.2020.116061>
- [42] Leerbeck, K., Bacher, P., Junker, R.G., Goranović, G., Corradi, O., Ebrahimi, R., Tveit, A., Madsen, H.: Short-term forecasting of CO₂ emission intensity in power grids by machine learning. *Applied Energy* **277**, 115527 (2020) <https://doi.org/10.1016/j.apenergy.2020.115527>
- [43] Lowry, G.: Day-ahead forecasting of grid carbon intensity in support of heating, ventilation and air-conditioning plant demand response decision-making to reduce

1979 carbon emissions. *Building Services Engineering Research and Technology* **39**(6),
1980 749–760 (2018) <https://doi.org/10.1177/0143624418774738>

1981

1982 [44] Maji, D., Shenoy, P., Sitaraman, R.K.: CarbonCast: Multi-day forecasting of grid
1983 carbon intensity. In: *Proceedings of the 9th ACM International Conference on*
1984 *Systems for Energy-Efficient Buildings, Cities, and Transportation*, pp. 198–207.
1985 ACM, New York, NY, USA (2022). <https://doi.org/10.1145/3563357.3564079>

1986

1987 [45] National Energy System Operator (NESO): Carbon Intensity API. [https://](https://carbonintensity.org.uk/)
1988 carbonintensity.org.uk/ and <https://api.carbonintensity.org.uk/>. [dataset]. Devel-
1989 oped by NESO in partnership with Environmental Defense Fund Europe, University
1990 of Oxford Department of Computer Science and WWF; Great Britain national
1991 and regional (14 regions) carbon intensity forecasts and estimates; accessed 29
1992 September 2026 (2026)

1993

1994 [46] Hawkes, A.D.: Estimating marginal CO₂ emissions rates for national electricity
1995 systems. *Energy Policy* **38**(10), 5977–5987 (2010) [https://doi.org/10.1016/j.enpol.](https://doi.org/10.1016/j.enpol.2010.05.053)
1996 [2010.05.053](https://doi.org/10.1016/j.enpol.2010.05.053)

1997

1998 [47] Siler-Evans, K., Azevedo, I.L., Morgan, M.G.: Marginal emissions factors for the
1999 U.S. electricity system. *Environmental Science & Technology* **46**(9), 4742–4748
2000 (2012) <https://doi.org/10.1021/es300145v>

2001

2002 [48] Vovk, V., Gammerman, A., Shafer, G.: *Algorithmic Learning in a Random World*.
2003 Springer, New York, NY, USA (2005). <https://doi.org/10.1007/b106715>

2004

2005 [49] Lei, J., G’Sell, M., Rinaldo, A., Tibshirani, R.J., Wasserman, L.: Distribution-Free
2006 Predictive Inference for Regression. *Journal of the American Statistical Association*
2007 **113**(523), 1094–1111 (2018) <https://doi.org/10.1080/01621459.2017.1307116>

2008

2009 [50] Angelopoulos, A.N., Bates, S.: Conformal Prediction: A Gentle Introduction.
2010 *Foundations and Trends in Machine Learning* **16**(4), 494–591 (2023) [https://doi.](https://doi.org/10.1561/22000000101)
2011 [org/10.1561/22000000101](https://doi.org/10.1561/22000000101)

2012

2013 [51] Romano, Y., Patterson, E., Candès, E.: Conformalized Quantile Regression. In:
2014 *Advances in Neural Information Processing Systems*, vol. 32 (2019)

2015

2016 [52] Stankeviciute, K., Alaa, A.M., Schaar, M.: Conformal Time-series Forecasting. In:
2017 *Advances in Neural Information Processing Systems*, vol. 34, pp. 6216–6228 (2021)

2018

2019 [53] Xu, C., Xie, Y.: Conformal Prediction for Time Series. *IEEE Transactions on*
2020 *Pattern Analysis and Machine Intelligence* **45**(10), 11575–11587 (2023) [https:](https://doi.org/10.1109/tpami.2023.3272339)
2021 [//doi.org/10.1109/tpami.2023.3272339](https://doi.org/10.1109/tpami.2023.3272339)

2022

2023 [54] Gibbs, I., Candès, E.: Adaptive Conformal Inference Under Distribution Shift. In:
2024 *Advances in Neural Information Processing Systems*, vol. 34, pp. 1660–1672 (2021)

- [55] Smith, J.E., Winkler, R.L.: The Optimizer’s Curse: Skepticism and Postdecision Surprise in Decision Analysis. *Management Science* **52**(3), 311–322 (2006) <https://doi.org/10.1287/mnsc.1050.0451>
- [56] Ben-Tal, A., Nemirovski, A.: Robust solutions of Linear Programming problems contaminated with uncertain data. *Mathematical Programming* **88**(3), 411–424 (2000) <https://doi.org/10.1007/pl00011380>
- [57] Bertsimas, D., Sim, M.: The Price of Robustness. *Operations Research* **52**(1), 35–53 (2004) <https://doi.org/10.1287/opre.1030.0065>
- [58] Google: OR-Tools. <https://developers.google.com/optimization/>. Version 9.15.6755, accessed 29 September 2026 (2026)
- [59] Friedman, J.H.: Greedy function approximation: A gradient boosting machine. *The Annals of Statistics* **29**(5), 1189–1232 (2001) <https://doi.org/10.1214/aos/1013203451>
- [60] Ben Taieb, S., Bontempi, G., Atiya, A.F., Sorjamaa, A.: A review and comparison of strategies for multi-step ahead time series forecasting based on the NN5 forecasting competition. *Expert Systems with Applications* **39**(8), 7067–7083 (2012) <https://doi.org/10.1016/j.eswa.2012.01.039>
- [61] Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., Duchesnay, É.: Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research* **12**(85), 2825–2830 (2011)
- [62] Aslan, J., Mayers, K., Koomey, J.G., France, C.: Electricity Intensity of Internet Data Transmission: Untangling the Estimates. *Journal of Industrial Ecology* **22**(4), 785–798 (2018) <https://doi.org/10.1111/jiec.12630>
- [63] Alibaba Group: Alibaba cluster trace v2018 (batch_task). http://clusterdata2018pubcn.oss-cn-beijing.aliyuncs.com/batch_task.tar.gz. [dataset]. Accessed 29 September 2026 (2018)
- [64] Guo, J., Chang, Z., Wang, S., Ding, H., Feng, Y., Mao, L., Bao, Y.: Who limits the resource efficiency of my datacenter: An analysis of Alibaba datacenter traces. In: *Proceedings of the International Symposium on Quality of Service (IWQoS)*, pp. 1–10. ACM, New York, NY, USA (2019). <https://doi.org/10.1145/3326285.3329074>
- [65] Künsch, H.R.: The jackknife and the bootstrap for general stationary observations. *The Annals of Statistics* **17**(3), 1217–1241 (1989) <https://doi.org/10.1214/aos/1176347265>

2071 [66] World Resources Institute: GHG Protocol Scope 2 Guidance: An amendment to
2072 the GHG Protocol Corporate Standard. <https://ghgprotocol.org/scope-2-guidance>.
2073 Accessed 29 September 2026 (2015)
2074
2075 [67] Shapley, L.S.: A value for n-person games. In: Contributions to the Theory of
2076 Games, Volume II, pp. 307–318. Princeton University Press, Princeton, NJ, USA
2077 (1953). <https://doi.org/10.1515/9781400881970-018>
2078
2079 [68] Diebold, F.X., Mariano, R.S.: Comparing predictive accuracy. Journal of Business
2080 & Economic Statistics **13**(3), 253–263 (1995) [https://doi.org/10.1080/07350015.](https://doi.org/10.1080/07350015.1995.10524599)
2081 [1995.10524599](https://doi.org/10.1080/07350015.1995.10524599)
2082
2083
2084
2085
2086
2087
2088
2089
2090
2091
2092
2093
2094
2095
2096
2097
2098
2099
2100
2101
2102
2103
2104
2105
2106
2107
2108
2109
2110
2111
2112
2113
2114
2115
2116